

A History of Useless Mathematicians

Simone Severini

Tell me the truth: true that it doesn't exist?

Alessandro Bergonzoni, *L'amorte*

PROLOGUE

At least twenty years ago

I was passing an apathetic night at the department. The room sat in a dense half-light, broken by two sources or maybe transmitters of photons. One of these was the laptop screen. Absurd to recall how a good part of my existence depended on the rhythms of that machine. It was a fixed presence. I knew its slownesses, its occasional tantrums. I knew which keys required a firmer press, there, where the edge had flaked away. I wrote, I worked, I kept in touch with friends and family. More than an object, it was a portal and an appendage. It was gray, with a thin black rim, while the keys, in that combination of green and yellow, gave it an almost sporty air, however faded by use. The “v” key, for example, was so worn down that it displayed a white crater at its center. That detail reminded me of the shiny snout of the Porcellino, the bronze boar fountain near the Ponte Vecchio. They say it brings luck. Not that the “v” had ever brought me any particular luck, let’s be clear, but I had no data to rule it out entirely. Sure, “e” is the most used letter in English and the space bar sits perpetually under the right thumb. So why the “v”? Together with “ctrl,” it is what you paste with. I guess I pasted a lot. But then, if I pasted so much, did I also copy just as much? The “c” key was almost intact. I am not sure what I pasted. Pasting without copying surely breaks some conservation law.

I thought about giving the machine a name. An adolescent idea. Naming an object is, after all, an act of idolatry, or maybe a slightly pathetic attempt at humanization. In my case, it was a clear signal. I spent the greater part of my time with the computer. Worse still, the thought lumped me in with pimply geeks. I mean the kind who can tell Snow Leopard from Affluent Albatross and work themselves into a fever over computing details that to me were entirely insignificant. In theory I should have felt part of their ranks. After all, I was doing a doctorate in mathematics. I had a kind of natural safe-conduct. And yet no. I was an infiltrator. Pimply, yes. Pale, that too. I lacked enthusiasm for technofetishism. I did not know how to program, except to cobble together, rarely, clumsy and fragmentary code.

So the computer remained simply “The Computer.” With its five years of service, it had lived through a geological era, on the timescale of electronic devices. I remember how it had already taken the place of a more ancient predecessor, in a cycle of successions going back to the dawn of time and that distant day #1 of my digital history. So “The Computer” was in every respect a noble title, handed down from generation to generation. The previous one had come to a tragic end. Burned. Burned alive, in fact, while it was still on. A detail that recalled Giordano Bruno. I remember the exact moment. I had found a CD-ROM inside a box of corn flakes, or *fiocchi di mais*, for those of us who still keep faith in the mother tongue. It contained a senseless game in which you had to throw little balls into moving baskets. I lost a good twenty minutes on it in the middle of the night. The computer rested on my legs, heating up until it glowed. I could feel the heat radiating. Then came a dull, vibrating buzz. The sound of an exasperated horsefly. Right after, a crescendo, an octave higher, like a tenor at the climax of an aria. *All'alba*. Then silence. Black screen, heat dissolved in an instant. “You, with your polite and placid murmur, you had analytically computed eigenvalues of every kind, and you abandoned me like this, crushed by a children’s video game.” There was no mourning. Some time later, without much thought, this new computer had taken its place.

I knew it well and this was a useful thing. I knew how many seconds it took to switch on, a strange number to measure, and yet sedimented in the subconscious. When it jammed, when it took an unrequested pause, I had the certainty of knowing how to fix the problem, or at least, through practice, I had accumulated enough empirical knowledge to predict its behavior. I had adapted to its rhythms and it to mine. My blood pressure synchronized with its noises, with the fits of the fan, with the electronic sighs by which it announced it was under strain. And so I had entrusted to it my notes, a collection of photographs, drafts of scientific articles that rarely made it past the second page, and even some short poems, written in moments of particular emotional energy. I kept it all inside a folder called “Everything.” A title without imagination, but effective. I did not realize how useless such a categorization was. If everything was inside “Everything,” then nothing was distinct from anything. The only purpose of that folder was to minimize the effort of moving the files onto an external drive, in case an

emergency backup was ever needed. In the end I trusted this machine, but only up to a point. It is always better that way.

So, on that night at the department, the screen was one of the two sources of light in the room. I was about to say “it was the first of the two sources of light,” but I see no need to create an order. The other was a square skylight in the ceiling, up on the left, above my shoulder. The color of the night beyond the glass was the color of fog on the highway, when the headlights bounce back at you and instead of lighting your way they blind you. Not far from my room stood the dome of a decommissioned astronomical observatory. I was thinking how impossible it was to see the stars in the city, lost in an age-old light pollution. I was thinking of Calvino’s *Marcovaldo*. The night beyond the skylight was not the night of blue and of the deep colors where ideas are born, like fluctuations of the vacuum. It was the weary night of taxis, of hospital lights, of planes on approach twenty kilometers out, so low and so loud they seemed like iron balls rolling across a wooden landing. It was the night of garbage trucks, of drunks, of signs still lit, of police sirens, of buses racing toward the end of the line. It must have been a quarter to two. That was the hour of nurses, of typesetters, and of the diehards of mathematics on the brink of failure. A nocturnal tribe I belonged to, immersed in a perfect solitude, between the computer and the skylight.

I was certainly tired, and yet wide awake from too much coffee. My dinner was coffee with two saggy biscuits. I was reading, or rather looking at, bits of articles on the arXiv, in the hope of finding, precooked, the proof of a conjecture of mine. I wanted to put to rest a *lemmetto*, that is, a little lemma, and move forward in the narration of a theorem for which I needed many *lemmetti* interlocked like the bricks of a wall. I was looking for help in the literature, hoping that someone more capable or more determined than me had already proved my *lemmetto*. I once read a proverb that went something like “If you want to eat a piece of bread you have two options. You can buy it at a bakery or make it yourself.” I would have liked to find it ready-made, my *lemmetto*, and move on. It did not even deserve to be called a lemma. But I needed it, the bastard. So I searched without luck, in the bakeries, the grocery stores, the supermarkets, the pantries, the dumpsters of mathematics.

“Signora, could I please have a hundred grams of porchetta. That half kilo of bread. That marinated eel? Pecorino and a few *sottolii*. There. Also the proof of this lemma.”

“Shall I wrap it for you?”

“That would be excellent.”

“Look, I’ll throw in these two corollaries here. Loose. Very fresh.”

“Signora, thank you. You have a heart of gold. But corollaries, they repeat on me.”

Someone must have written it down, that proof. A detail so obvious was bound to turn up. Once, twice, five times. Maybe the textbooks took it for granted precisely because of its absolute banality. I imagined all those minds of the past, bent under neon lights, candles, or oil lamps. There, at two in the morning, salvation depended entirely on me.

“A little help, give me one, won’t you? Ancestors, I am one of you, all things considered. Tell me where you put this proof, please. Or at least give me the strength to do it myself. O divine ones, omniprovers of mathematics.”

I had tried. Flour, water, yeast, and salt. Once in the oven, the dough failed its transformation into bread. I hoped for a fragrant crust, I got only an indigestible little brick. I changed method. I hoped for an airy, open crumb, I pulled out a little loaf of plasticine, dry outside and raw inside. The absence of that proof from the literature multiplied my humiliation. Its general omission meant one thing only. The authors had left it out of the books judging it too obvious, too self-evident, too immediate. I, on the contrary, kept groping in the dark. It is demoralizing to feel incapable, meant in the literal sense of lacking capacity, of being short on volume, like containers too small to hold the proof. She overflows, she exists, conceding herself to my wits only in her containable portion.

Defeated, I lowered the computer screen. I was left with the skylight as the only source of light. I left the room, headed home, with what little energy remained. My brain was working badly, fogged by fatigue and hunger. Going down the stairs, two nocturnal stories from my time as an intern came back to me. The body descended, memory climbed back up. A connection created by my subconscious. Events at least three years old.

I had recently arrived in a place totally new to me. The research institute was in a suburb of Vienna. I slept in a room of a newly built building. Long. Modern. A student residence. It had six floors. A white and pink parallelepiped. I had arrived before the start of classes and everyone was on vacation. The custodian pointed out to me a notable

fact. The building was completely empty, with me as its only tenant. It would fill up soon. The room smelled of new. One side of the mattress was still wrapped in cellophane. The ground-floor door let you into a lounge. There were plants. Ficus? You walked to the elevators. You went up three floors and found yourself in a long, aseptic corridor. There was never anyone. The bathroom of my room was a plastic box siliconed into a corner of the room. It contained the crapper, a sink, two hooks for hanging towels, and in a corner of the corner, the shower. Very efficient. Hot water in one second. From the window of my room you could see a more or less identical building. White and greenish. A pale green. I figured the residences were two: the North and the South. I was in the South. The first nights I closed the curtain just before getting into bed. Then I began leaving it open. With the curtain closed, I did not get the dawn light in the room, though the autumn morning light was feeble. I did not have to close the curtain to shield myself from prying eyes in the other building. The North residence was completely empty too. The only light visible in the windows of that building was mine. It was the light of my room reflected in the panes of a room in the North, I imagine, similar and mirror-reversed.

One evening a small miracle occurred, of the same order of magnitude as my lemma. Closing the curtain, looking out, I noticed a second light in the North. I tried to figure out whether it was a strange reflection due to a window open somewhere. The second light surprised me. And it surprised me more when I saw it again the following evening. Someone was sleeping in the North residence. A male me or a female me, antipodal to my room. Could this unexpected presence have some magical meaning, hidden from me? Had they left a light on? Driven by curiosity, I decided to use a simple method of verification. One evening, having made sure the light was there, I went down the stairs, crossed the outer courtyard, and entered the other building. I tried to work out at what point along the corridor the room of the light might be. I read the numbers on the doors. I was starting to feel like turning back, because it was hard to tell whether the light was on my own floor or another, above or below.

When I decided to leave, a door opened a few meters from me. Behind the door I discovered the room of Balazs, a Hungarian student. Balazs had opened the door because he had heard my footsteps. I told him I lived in the South residence and, in case he had noticed a light, it was mine. He smiled and invited me into his room to eat slices

of a cheese called ranolder, named after János Ranolder, historic bishop of the town of Veszprém, a fair number of kilometers from Lake Balaton. The mystery of the light was finally revealed. From that day on I felt in company and much more serene, reassured by the presence of Balazs in the North residence.

The second story tells of an event that took place a few hundred meters from the residence and in the same interval of time. My internship supervisor, a person of rare kindness, had assigned me a desk in a somewhat isolated room of the institute. It looked like a garret. I shared the room with some doctoral students. They were three, four years older than me. The institute was housed in a medieval castle with mighty walls. Inside, though, it was ultramodern, with glass, concrete, and minimal architecture. The rooms went in perfect accord with the shell of the building for inexplicable reasons of aesthetics. To reach my desk you had to climb a spiral staircase. You then found yourself on a mezzanine, where a glass wall separated the room from the stairs.

One night, late, I had stayed on alone. I guess when you have this habit you have this habit. The sensation of being the last to leave a place, any place at all, has always caused me a certain unease. Being last makes me anxious. That evening, specifically, fell on a Saturday. Downstairs there was a kitchenette. I got up from the desk in search of a glass of water.

At the door, the corner of my eye caught a movement. A blurred, and equally sharp, silhouette. It pointed in the opposite direction, heading straight into the room. It was barely five meters away. I spun around on the top step of the spiral staircase. I found absolute emptiness in every corner. My officemates had gone hours before. A trick of mirrors, I hypothesized for a millisecond. In the next millisecond a tremendous fear took over, the paralyzing kind. A ghost? I faced the descent very slowly, tilted forward, glued to the handrail to feel safe. Eyes planted downward, neck shrunk in. The image in the glass wall was my reflection. A logical, rational explanation. Like when I was a child. My father, in the kitchen, would ask me to go to the bedroom and fetch him a cloth handkerchief to blow his nose. I made a considerable effort to reach the bedroom hugging the baseboard and switching on every possible light. And the rule, at home, was not to leave lights on in empty rooms. I contorted myself, pressed my back against the wall, and sometimes broke into a run. I was terrified by the possibility of stretching my fingers toward the switch and realizing I had not flipped it while my arm was already

returning to me, and so finding myself in the dark by mistake. I had to solve an optimization problem to manage to turn off the lights on my way back to the kitchen while trying not to end up in the dark. And the fear of the dark grew heavy when I tried to fall asleep in bed. I was covered head to toe, keeping only one eye out, like a periscope, and one nostril, of course, because the air under the covers is unbreathable.

It is so interesting to observe how our psychology changes over the years and as a function of the seasons of life. Lately I sought out the dark to rest my eyes when I could not stand the computer anymore. I truly adored pitch darkness. Come to think of it, I believe I loved mathematics precisely to guarantee myself a safe refuge, immune to fear. The symbols you shape to your liking. The monsters reduce to complicated algebraic structures, entirely harmless in the dark. Frustration costs less than fear. Do not tell me that people like mathematics because they are curious. We love mathematics to decrease our fear.

Now I was going down the department stairs asking myself why the lights stayed on all night. A waste of electricity. The fear of the dark had been displaced by the more concrete and opaque fear of being left without a research contract the following year. I reached the hall on the ground floor, where they usually put on art exhibitions connected to science, mainly biology. I remember an exhibition on DNA. They had placed sculptures of metal or plastic along the sides of the hall. They depicted DNA. Some were a couple of meters tall. I remember how one of them fell during installation. The double helix had split down the middle into its two strands, a bit as happens in the replication process carried out by nature. They stuck the DNA back together with adhesive tape. Since it would not hold, they had to run a rope around the double helix, from the bottom up to the top of the sculpture. The DNA had become a triple helix, thereby reenacting the action of the *E. coli* recombinase RecA.

The hall looks out onto an inner courtyard, a springtime gathering spot for students on lunch break on the grass of the central flowerbed. At night, the neon hiccupped exactly as in horror films, promising at any moment the shadow of a big kitchen knife on the wall. Now the walls were adorned solely with the posters of a conference on superfluidity.

Walking from the stairs toward the door, on the left, I glimpsed a fox. Still, in the middle of the hall. Reddish. Bristly. Scrawny. Furry. A real creature, in flesh and bone. It

stared at me with two round, wide-open eyes. It had interrupted its retreat expressly to look at me, head turned backward. I was still too. Motionless. Such an encounter at the Department of Mathematics bordered on the absurd. Ears pricked. As a human being, I ought to feel a variety of emotions crossing paths with a fox, at night, suddenly, and in so unusual a place. Reactions strictly tied to the mood of the moment. The most obvious options include a certain tension, this being a wild and potentially aggressive animal, given also the exceptional nature of the situation. The reason for its presence verged on the banal. It had found the door open, exactly like the proverbial dog in the church.

So I was not afraid of the fox. We have been told that fear has a cognitive function, that it serves to channel our senses into solving an immediate problem on which survival depends. I felt only sadness, because that living being was looking at me with a surprise identical to mine, almost as if it were asking, “Go to bed, why don’t you? You are wasting your time. You are not cut out for mathematics.” All at once, in its animal being, the eyes of the fox showed themselves compassionate. I managed to read in them the weakest form of empathy. It was as if it understood me.

“You are here alone in search of a role. All human life is pure narrative. Mathematics simply represents one of the clearest narrations. Go home. And tomorrow morning, tell your friends about this encounter. There are many ways to tell a story. Not just mathematics.”

In an instant the fox moved its head and went skipping off. A gesture to display its lack of interest, right after dishing out in front of me the truth about my condition. I raised my arm to wave goodbye. I lowered my head as a sign of respect. It vanished in silence toward the Department of Physics.

Things demand a story in order to have meaning. Mathematics itself is the telling of detective novels in which the culprit has to come out. Even truth, yes, it is a physical resource possibly consumable by the actions that it enables. But if it is not set inside the story, it is of no use at all.

I went outside into the courtyard to sit on a metal bench. It was freezing. Under a sky in every way identical to the skylight’s, I found myself at peace with my smallness. At some random point a star was shining. Or maybe it was a piece of technological debris in orbit. I smiled, with that dry detachment of someone who has taken the measure of their limits.

CHAPTER ONE

Nine years ago, late evening

I have incoherent thoughts. My brain has surely been rewired by too much use of screens, cell phones, little messages, and a thumb by now prehensile, like the tail of a howler monkey. Like everyone, for that matter. Electronic objects have trained me to interrupt any line of reasoning before it can even take shape. Or maybe it is simply my nature that has led me anyway, inevitably, to be a chronic scatterbrain. I am talking about a condition which, come to think of it, was already showing its symptoms when I was a child. Maybe this dispersion defines me. Maybe more than a defect it is a mode of operation.

How did SeneQCa Inc. begin? It was one of those days when the universe punishes civil society for some sin of the ancestors. It was raining with ferocity. Big gray clouds bulged toward us like cysts. The circumstance took the shape of an inauspicious omen, an oblique shadow mixed into what, with our usual intellectual zeal, we call the celebration of complexity in physics. They are called, precisely, “complex systems.” It all depends on the perspective of the individual, I suppose.

We were in the vicinity of Hackney Road, in a pub, the kind with sticky carpet. I was sipping a lousy pint, lukewarm and bitter. I watched a breath of condensation form on the glass, conjecturing a future acidity of the stomach. I tried a couple of times to make the glass sing by rubbing the rim. Nothing. You need a wine glass. The beer mug is too rigid and requires too much energy to be deformed by the simple rubbing of a finger.

London was a good place: there was a high concentration of talent produced by good universities; it was easy to found a company; and capital of global provenance moved through London. London was a bad place. Too expensive for our pockets; high social pressure; sad climate. People arrived, worked, and left; or, when they started a family, they ended up in suburbs so remote that to go visit them you had to look up the street names on ancient, nearly indecipherable maps, unearthed only on the stalls of junk dealers. Hyperbole aside, none of us lived in the center.

A fund, already busy investing in ideas coming out of our university, had vaguely promised support on condition that we put together something resembling a company. They pronounced: “You need a commercial vehicle.” The phrase still makes me think of a small truck. I imagined back then a group of scruffy academics, crammed into a van, selling algorithms instead of sandwiches. I smile. We had to get a move on, though, faster than my mind could process the transition from proving theorems to seeking profit. Within a few weeks, we registered the startup without the slightest understanding of the future. And I am not claiming that the future must necessarily be understood, or that this is feasible. The lawyers closed the investment in record time, their efficiency in sharp contrast with the languid rhythm of the academic life I was used to.

According to certain experts, a project has no right to exist unless it is first crucified on the four quadrants of an Eisenhower matrix: urgent and important, urgent and not important, not very urgent but important, neither urgent nor important. Four quadrants. That is why they are called quadrants. I have always thought that, had Eisenhower been a lover of mathematics, he would have opted for something more complicated, like a Latin square or a Leech lattice, mysterious objects of our bestiary.

Quadrants have always annoyed me for two reasons. First, I prefer to go on instinct, following small, immediate emotional rewards, without pressure. Second, in the continuum of reality I find it hard to define distinct sets of things. From quadrants you pass to octants and so on up to orthants and an exponential impasse. Before you know it, you are dealing with some strange multidimensional object and the whole point of making the workflow simpler is lost. Our Eisenhower matrix had a single quadrant, with “dunno” written in it. Paradoxically this made me feel more at ease. If you have no categories, you have no distinct emotions to attach to those categories either. The absence of strategy is liberating. On the other side, terrifying, yes. It is like jumping off a cliff and building your wings as you fall, a metaphor used very often in the world of technology, maybe too often, something I would learn only later.

More than the imminent investment, at the pub we were celebrating the possibilities opened by the void. This because our business plan was a morula. All we had was a presentation, a pitch deck, not very convincing, which nonetheless succeeded, given perhaps the investor’s haste, owed in part to a particular confidence endemic to the

mood of the financial markets. Thinking back, it was a decent presentation, visually tidy, as if in dissonance with our mental disorder.

The idea was to build a supercomputer to do something spectacular, where the how and the what remained to be determined. Put together processors built by others? Build a new processor ourselves? Totally at random. We had in our pockets a few secret ingredients, minor ones though, and the support, both technical and moral, of Albo Gilmor, the great scientist. From the outside this was read as a guarantee of quality. He had been our doctoral advisor. I remember him still, eyes wide with enthusiasm, gesticulating animatedly as he tried to convince us that our secret ingredients would revolutionize computing. And as he explained he launched sprays of saliva that landed on us, forcing me to take a discreet step to the side. Gilmor was a schoolmistress generous with compliments as she pats children's heads after they have drawn a little man in felt-tip. Everything had to be interpreted in the right key.

Naturally, the success of it all remained tied to the hypothesis of unshakable perfection, to the occurrence of propitious occasions, and to a benevolent fate spinning in our favor. Hence, the probability was almost nil. The secret ingredients, moreover, described how to build a hybrid machine, made of different hardware designed to solve different problems and connected through a network with very low latencies. Not even a new idea, really, but Gilmor had a way of making even recycled concepts revolutionary, with his thick eyebrows and halo of gravitas. Now I know. His enthusiasm was genuine, though, moved by the awareness of how much cheaper in energy it is to be positive than to openly admit to being losers.

But it is well known, in the trade at least, that whoever founds a technology company has a dream afflicted with gigantism. There are those who tell the dream in a low voice for fear of the envy of the gods, and those who put it under their feet like a flying carpet. Such a founder moves around permanently thirty centimeters above the floor, wears turtlenecks or pleather jackets, like epic heroes of industry, and begins to claim to be a visionary. We know, however, that visionary people come into existence rarely, while there is an abundance of those who prattle about the giant dream as if it had already come true, or, worse still, as if it were the only sensible dream, while everyone else's dreams are pathetic. For the record, I always wore the same blue sweater, with commemorative conference T-shirts underneath.

“The only way to keep the dream alive is to dream it.” Nice phrase. When that dream begins to lie, you must lie at all costs, independently of the state of the universe, even when you realize the uselessness or the unfeasibility of your technology. Albo, as spiritual guide, already back then, in his words “a relic of the past,” admonished us with a lapidary “You wanted the bicycle. Now pedal,” when we received the term sheet, that is, the draft agreement at the base of negotiations with the fund.

At this point, all we needed was a name for the company, something as useless as it was essential. To find names, we dug into Greek and Norse mythology, a bridge between an arcane past and a mystic future, as in the respectable works of science fiction.

Giving the right name to something with a purpose, even if it does not exist in the physical world, almost serves to bring it into existence. Ontologically, your existence is enriched by a label called a name. Or at least this was the narrative while we scribbled increasingly ridiculous words. We considered option after option, writing them on a paper napkin. That is what they do in the movies. Nerdy, but with style. We needed a Q, given the connection with quantum physics. This supercomputer was supposed to have something to do with quantum physics, a topic everyone was talking about, from academics originally interested in physics of the most mutually distant kinds, to self-styled experts on social channels, down to the people you meet at the supermarket in front of the yogurt fridge, or worse, over a bunch of leeks. We too had to ride the wave of the hype. Fashion lifts you like a tide.

Here are the names, the ones I remember. ArQuonauts. No. Queson and the ArQuonauts. No. Good for an alternative rock band but not for an enterprise. Qretheus. Sounds a bit like cretin. ArQimedes. Qiron. Qsidero. OrQomenus. MaQuesian. LolQus. No. OrQor. No. They all sound like monsters in role-playing games. HarQoh. Unpronounceable in any language. Albo had already proposed Osvaldo, the name of an old dog of his, dead for decades. We did not approve it. Bad omen. Scaramanzia. Then came SeneQCa Inc. Beautiful. A philosopher, hence an indisputably wise person.

Toward a quarter to eleven, before the last-orders bell, Sandra Bianchi, Melvin Chung, myself, and two other people irrelevant to the economy of this story converged and landed on SeneQCa Inc. It became the name of the company and of the technology produced by the company, that is, the supercomputer.

Sandra's eyes lit up at the name. Her usual pragmatic attitude softened for a moment. Melvin nodded in approval, already imagining the logo on business cards and billboards. As for me, I was merely relieved that we had not called ourselves something even more ridiculous. Days later I said something like "Seneca was forced into suicide by his pupil, Nero. Bad luck?" Within one second, Gilmor came back with "...facing death with imperturbable serenity." We changed the subject.

Melvin at a certain point became chief executive. I remember an interview of his appearing somewhere on the web. He said: "We founded SeneQCa Inc. to change the history of computing." Are we sure? To tell the truth, we founded SeneQCa for three reasons: first, some investors were giving us money to do it; second, our miserable university salaries barely kept us from sleeping in the street; and third, sometimes in life we make decisions and cannot explain to ourselves why. In light of which, sorry to say it, the dream of changing the world is much more attractive when you are not surviving on spaghetti aglio e olio, despite thirty years of academic curriculum. It was a habit and a necessity of mine at the time, given that my salary sat at the poverty line, as established by some national statistic. I appreciated the carbohydrates, while I exuded love for the future. It seemed noble and important.

Melvin, for his part, had a mastodontic ambition: his father was a professor in Zurich and his mother was a high-level manager at a bank. Melvin's mother had also won a medal at the Olympics. Equestrian. Sandra truly believed in the project. I wrote useless articles seven days a week and was looking for a blessed escape route from an academic life that by now stank of mold. The idea of building a supercomputer seemed like a good option and at that moment it was the only one. You drink or you drown. At the time I did not know that on the exact day you start talking about money, problems of every kind arrive. The primary problem was my inability to talk about money.

Had I been more open to change, I would now have a villa somewhere in Malibu, a cottage in Gstaad, a third of a private plane to take me to Burning Man, and maybe a Cirneco dell'Etna, called Max, Rocky, or Sparx. More simply, I would have called him Caianiello, the great founder of the Neapolitan school of cybernetics, or Gregorio Ricci-Curbastro, the inventor of tensor calculus.

At the end of the day, my shares in the startup were proportionally few, but decent, considering also that I had kept an academic position and therefore, at least at the

beginning, was not in the company full time. A portion of my shares was used to satisfy the interest of some latecomer investor when we later closed various capital raises. I thought they were doing me a favor by letting me sell my shares privately. What looked to me like solid gold was, in reality, a trifle for whoever bought them, in proportion to the depth of their portfolios.

If I look at this experience, by now from a distance, what have I learned? First of all you need a vision. It is not enough to have a potentially good product, a valid plan, or a capable team. Vision is the glue, that almost invisible force for holding everything together and giving people the desire to keep going, even when things turn bad. Without a vision, and without someone to embody it, the engine is missing. You proceed by inertia, until you stop. That is why you need, sometimes, a prophet. Not in the mystical sense of the term, but someone capable of generating a collective cognitive alignment, pushing people to believe in the project. In a startup, convincing the people who work there is almost more important than convincing the people who invest. It is a matter of creating a fan base inside the company. And without a fan base in love with the team, the game cannot go on. This I learned on my own skin. Melvin was not a prophet. Me? As if.

The second lesson: in a startup, the narrative comes before the product; first you tell the story, then you build. This is something that still leaves me disoriented. I was born and raised in science, where romanticism is almost everything and there has always been a mix of idealism and a certain modesty about publicizing one's results, at least historically, perhaps for fear of the judgment of one's community. "The results speak for themselves!" I had been trained to work on the formalization of problems. In my head, the narrative was a consequence of the results, almost superfluous, and certainly not a premise.

I spent years feeling my way, trying to solve problems guided by an inner compass, or better, a dowsing rod pointed at the solvable: "let's solve this problem because it looks tractable," or, "let's fix this part, and leave the rest to posterity." Running a startup is another world. Raising capital means convincing people with money in their pockets. Presenting plans or projections is useful. What you certainly need is a strong story. A story not merely credible, but one that feeds the irrational conviction that the investors will get their investment back multiplied by ten. Moreover, you need something beyond

trust: an almost religious belief in the vision and in future success. It is like selling a promise, a still nonexistent dream that all the parties involved, employees included, must know how to imagine. Hence the usefulness of the prophet.

All of this, put together, was for me the real mental leap. Nine years have passed since that evening at the pub. Since then I have consumed, rough estimate, something like a thousand liters of coffee. A remarkable figure, I would say. Beer, on the other hand, like that evening in London, almost none: I have the impression it sets off my reflux, or GERD. My present-day reflux for quantum physics I call QERD.

Three and a half years ago, afternoon

The opening sentences of certain articles in the mathematical literature are always the most beautiful, thought Albo Gilmor. *Let G be a graph. Let S be a set. Let H be a subgroup of a finite group. Let M be a matrix. Let X be a random variable. Let f be a function. Let V be a vector space.* And one could go on and on, with multiple cases across the whole alphabet, since the list of possibilities is long and baroque.

These sentences have remarkable value because they define the context. They do so essentially. They create it. They draw a sharp border between abstract worlds, thus partly unknown but controllable, and human life, that is, a realm swarming with meanings and uncertainties of every kind.

While they build that context, mathematicians play God, in a certain sense, wearing the robes of a demiurge with paper and pen. And it must be said that when you are God, you feel fairly safe. Maximally safe, in fact. Such sentences also give a subtle pleasure in the writing: the pleasure of manufacturing a small universe from scratch, similar to an aquarium.

Granted all this, *Fiat lux*, “let there be light,” is another question entirely. It refers to that physical world, made of matter, which properly includes human life, and precisely for this reason is extraordinarily complex. The construction of the physical world, one could argue without much risk, precedes our existence and is independent of us. Light is not a mathematical object, at first sight. It was handed to us by a supra-mathematical divinity, who mockingly forgot to leave us the instruction booklet, which for hundreds of

years we have been trying to reconstruct through that bizarre activity called “scientific research.”

Still, the imperative “let X be” raises intriguing questions. Does every abstract object correspond to an entity in the physical world? And does every abstract truth find a parallel in some material truth? Improbable, even if it is our physical world that suggests, however remotely, our abstract truths. It is inevitable, given that we live immersed in physicality.

While he was entirely lost in these disconnected reflections, Albo Gilmor brought two truths into focus: that his colleague was visibly late for coffee, and that his life had by now passed eighty years. Two distant thoughts, yet connected by a thin thread made of time, with its double nature. On one side, it marked the small daily rituals, like a shared coffee; on the other, it was spread across the immensity of an entire life, if you are lucky, once it is accepted that living long is a good thing.

Heading toward the bistrot kitchen, the waiter stopped for a second to ask whether the Professor wished for dessert at the end of his lunch. Chocolate mousse. Gilmor raised his left eyebrow, pronouncing a baritone but undecided “*Pas encore*” with the intention of saying “In a little while.”

He folded his napkin with a rigid motion, as if executing an algorithmic procedure for producing origami. It was a deliberate act charged with an almost symbolic weight. It was the kind of gesture performed by someone who has spent much time contemplating the hidden order of things, looking for patterns even where there are none, or where they are so devious as to escape common perception. His hands moved with a studied exactness, as if to compensate for an inner uncertainty. Or it was just the gesture of an old man.

He peered out the window at the colors and warmth of another late spring on the avenue. He thought about the time needed to go by metro from the Panthéon area to the 16eme ARR. And he saw himself, on foot along Rue Saint-Jacques fifty years earlier, with books under his arm wrapped in a page of newspaper. There was a sullen bookseller. He never addressed a word to you, but he knew life, death, and miracles of French mathematicians like the great Galois and Bourbaki.

While he brought his eyes back to the napkin, to see whether he had folded it well into a square, he wondered what the proof of the theorem consisted of. What the crucial

step was for achieving that phase transition between darkness and the true. His “young” collaborator was arriving precisely to help him get to the bottom of it.

It was an extremely important result, that theorem. Everyone was talking about it, even the international dailies, in a media storm owed partly to the lack of other notable stories. Despite this, he could not find the mental energy or the willpower to read the article.

It must be said that in the world of scientific communication, the fact that something is talked about does not necessarily imply that the something is significant. Nevertheless, the marketing of science by now permeates every link of the assembly line, from the distribution of funds to the choice of research topics.

Gilmor was a natural attractor in the discussions around the theorem. It was described to him repeatedly, insistently. Sometimes with pedantry or with the desire to make a good impression, in equal amounts. He was, after all, Albo Gilmor, noted scientist, and godfather of an entire, and new, generation of mathematicians, ready to use computers to try to prove theorems. Once, the term would have been “computational mathematicians.” Now one spoke of verification and inference, and less of computation, less of calculations. The calculations, the old-fashioned kind, were handled by the machines. Mathematicians had become developers, enablers, inferencers, conjecturers, informal thinkers, formal designers. “Deductionistas,” with a linguistic closeness to what the followers of high fashion might say: fashionistas.

Gilmor was now the venerable prince of the deductionists, people who prove theorems in cahoots with the machines. They use the machines as their assistants. But the machines do most of the work, the painstaking, cognitively onerous, inhuman part. And us? Parked, to think bigger things.

The article was a tour de force through the meanders of number theory and a potpourri of mathematics abstruse because at first sight unconnected. It used difficult conceptual machinery to put to rest a question that had defeated hordes of relentless human solvers armed only with gray matter and the patience of saints.

Obviously, it was now a different game precisely because of the direct use of machines. The research landscape had transformed. SeneQCa, the supercomputer, had assembled the fragments of a formidable proof. Not even the best trained minds could fully decipher it.

A clean victory of the machine team over the old team of biological brains. Or better, a clean victory of the human team capable of working together with the machines, against the conservatives, snobbish or fearful of being overtaken. Mathematical symbiosis. It was one of Gilmor's favorite metaphors. A chess grandmaster beaten by a washing machine? No. A work of years, meant to build a giant archive of theorems on which, through the machines' capacities, we can draw with minute care and identify imperceptible connections, almost hidden behind semantics against which intuition stands there dazed. A microscope for archivists.

"Congratulations," he heard said to him, even too often. He navigated waters where, when people congratulate you, they sometimes do it for personal convenience, for political visibility, or because forced by invisible hierarchies, while sharpening knives with their hands behind their backs.

Gilmor was originally the principal theorist behind SeneQCa, the project at the summit of a life. He himself defined how the various parts of the machine work together. He formulated protocols for transporting only the useful information during the process of computation. The machine chopped it up and compressed it somehow. He reused results discovered in the past and fortuitously useful. It was he who signed the many patents. Some thirty, of which about a third with Sandra Bianchi. With age, however, a certain intellectual tiredness also made its way in. He refused to speak with journalists and preferred to continue with the usual routine.

"When old age arrives, they come to you to ask favors in the human context, not the technical one. It helps to remain relevant on the basis of the social fabric built over the years. From the technical point of view, they consider you inert. Worse, actually: mathematically dead, as Erdős would have said, referring to those who no longer prove theorems. I am dead, this we already knew. SeneQCa has buried me." They were cutting words, the fruit of a personal reflection carried to excess.

His collaborator, an energetic person and once his doctoral student, back when one still had to suffer in the solitude of unwritten proofs, would save him today from the ordeal of reading the article from the first page to the last. He trusted him. What is more, talking with him was a way to dodge the self embarrassment felt every time he preferred to take a walk in the park instead of diving into mathematical writings. The sense of guilt sticks to you.

It was not only a question of age. It was an emotionally expensive process. Exhausting. He missed the desire, that drive which once devoured him, forcing him to gut every problem down to its essence, at the cost of sacrificing sleep. On one side he now felt at peace. At least that. It was the peace of someone who no longer has to prove anything, an unintended play on words, and of someone who can observe the mathematical world from afar the way one looks at an old battlefield. On the other side, he suffered a certain sadness, as if he had lost a demanding, neurotic, but passionate lover, whose memory was now little more than a black and white photograph. He remembered how procrastinating on difficult tasks is a direct method for avoiding unwanted emotions. The difficulty lies equally in determining which emotions are the wanted ones.

Happiness is a state of mind and not a function of material facts. It can be taught and even imagined, molded and engineered. Standard happiness is prescribed by what society believes happiness to be. The social definition proposes a universal model.

While he reviewed his relationship with his daughter, Angela, and their emotional distance, a question about the maximum of happiness made room for itself, silently, among his thoughts. “Professor Gilmor, what was the happiest day of your life?”

Assume you can quantify anything that has happened and anything you are. Assume you can compute a function with many variables. Very many. Quantify prestige, money, knowledge, affections, physical and mental wellbeing, value created for others, time gained or lost, influence exercised, environmental impact, reputation, personal satisfaction, future possibilities, social recognition, emotional stability, decisional autonomy, sense of purpose, intellectual freedom, attention received, risks avoided or taken, integrity maintained or compromised, justice promoted or hindered, responsibility, fairness in relationships, coherence between means and ends, respect for the dignity of others, damages repaired, transparency of actions, tension toward the common good.

What else? We are not talking about the external world, only about you as a person. Physical pleasure? Do we want to talk about an idea of global happiness? No, we want to focus on a technically individualistic and utilitarian perspective. Set morality aside, if you can. Quantify also the other dozens and dozens of parameters that right now we are not even able to define and put in a list. Evaluate all of it. Maybe take an average across

all these parameters and determine your degree of happiness. Are we sure it can be done? It cannot be done. Maybe one must turn one's attention to unrepeatable, utterly singular moments, like births? Maybe. In a birth there are stronger emotions, like fear. Emotions difficult to separate or even describe. Weddings? Ridiculous.

Happiness cannot be computed, just as one cannot compute the shortest program in Kolmogorov's theory, according to which the complexity of an object, a number, a string of bits, an idea, is given by the length of the shortest computer program that generates it. But precisely this program, the most essential and synthetic, is in general undecidable: no method exists that allows one to find it with certainty. In the same way, no definitive formula exists for measuring happiness.

Just so, Albo Gilmor was not able to determine the happiest day of his life. He found a few local maxima, but the global maximum escaped him. He thought of the careers alternative to his own that he never had a chance to consider seriously. He thought of the Mediterranean scrubland. The smile of his granddaughter, that is, Angela's daughter, appeared before him. Tiny white teeth between little strawberry lips.

"Grandpa, give me that candy," on the couch, she would whisper in his ear, knowing that in the kitchen there was a plastic jar with soft little orange candies inside. They were a kind of sugar coated jelly. And thinking of his granddaughter, he could no longer understand whether he was building in his mind things that had never happened or re-imagining reality. Time passes so quickly. "Grandpa, I want chocolate. Grandpa, will you bring me a lollipop? Please, grandpa." He coughed into his fist and with a nervous jerk undid the napkin, dismantling the origami.

He saw himself small, together with his parents. Rare but vivid scenes. They were on a meadow. He saw his father save a ball destined for the brambles. He wore a shirt from another era. He saw his mother wearing a long cotton skirt. Colorful. Everything seemed to glitter with summer. The green of the meadow was a surreal green. Emerald green. He saw his father smile with sincerity and his mother sitting on a red and white checkered cloth. These memories, perhaps, were the closest to his understanding of happiness.

A little later he watched the curve of happiness push lower, away from its upper bound. He thought for a fleeting instant of his wife, gone now for years. Technology or no technology, sometimes it changes zero. You love a person. The illness arrives, and so

long. Go ahead with your data analysis, DNA screening, preventive and personalized medicine. In the end, we are always, all of us, hanging by a thread.

Did he remember the sad and dramatic days better? He recalled details after having forgotten the actual reasons for each tragedy. He saw his father again, catching the ball with his hands. Turning, his mother was smiling too.

“Professor Gilmor, what was the happiest day of your life? I have been happy when I felt the happiness of others. I have been euphoric, more than anything. Happy, little. Very little.”

“Then why did you not work to give others this happiness?”

“Because I am afraid of my own emotions. Happiness has a very bitter aftertaste. Perhaps out of scaramanzia, too, I did not want to be happy. Being happy brings bad luck. You are happy for an hour and an eternity of struggle follows.”

At that instant, I entered the restaurant. I was sweaty, and I felt a cold interspace between the skin of my chest and my sweater. I had on a blue wool pullover, with two moth holes, which had once belonged to my grandfather. I have washed it so many times as to turn it into a bathmat, or an exoskeleton of felt.

I was the young collaborator. Young in a manner of speaking, and in any case relative to Gilmor, I was. Without a doubt. I had a scraggly white beard at the sides of my chin and a circle of alopecia, completely bare, under one cheekbone. It was hot and I deserved to sweat. I had forgotten to run the washing machine and had no T-shirts to wear. The backpack on my shoulders made the situation worse. But I had to carry my computer, notes, et cetera. To remedy this, that afternoon I bought, in a souvenir shop, a white T-shirt with *I love Paris* written on the front. It was true.

Gilmor sat on a small sofa, at a table for four, near the window. He spent every summer in Paris, where he was the guest of I don't know which university. He wore a light blue polo shirt, completely unbuttoned. Three white buttons. High forehead, fixed since forever in an expression of concern. White hair. Perfect teeth and no glasses. He smiled often, lowering his eyelids slightly. When he did not smile, he assumed an expression as if of pain, but of a stoic pain, bearable to him. A man some fifteen centimeters taller than I.

Our first meeting in person was in Zakopane, Poland, decades in the past, during a winter school in computer science. It is a place of which I remember the mountains and

the pierogi. We ate smalec, a traditional Polish spread made from lard, on crisp bread. I was in front of a slice of bread when I learned that Gilmor was an invented surname, derived from an Eastern European surname like Morawski or Holimorovsky, changed by grandparents or great-grandparents more than a century and a half ago. I wore my hair to my shoulders. Extremely greasy. I had both the hormone and the zeal of the mathematician. He took me as a doctoral student. I spent sleepless nights in the attempt, often vain, to prove some theoremlet, and spent the following days, drowsy and confused, daydreaming nonexistent proofs, which seemed to amuse themselves by remaining unreachable to me in their ontological state, far from my grasp. I too, like Gilmor, had an infernal hunger for truth. But if our field had been the high jump, he was Sotomayor, while I was the one who won the student championships only because nobody else was competing. And yet I survived.

I have always felt Gilmor to be a sincere person; one who does not try to embellish his speech to tend some personal interest, whether psychological or worldly. He lacked sweeteners. He was a luminary. He literally illuminated the dark corners of proofs. Unfortunately he intimidated everyone who did not know him well. He did not intimidate his students. He had patience with us and always a kind word.

Joking, I told him once, “They have referential awe, not reverential,” where “referential” meant I don’t know quite what, but it sounded good. He laughed and resold the joke, because referential is a computer science term used in the context of relational databases. I don’t know, precisely, whether other people’s awe influenced him, but after two clean minutes of conversation you noticed what a fragile, romantic spirit he was. He was a misfit in the academic temple, where one must also learn to look at things with distrust. Over the years I discovered that Gilmor was a misfit everywhere, inside and outside academia. Mathematical savoir-faire does not automatically translate into that of non-mathematical life.

“I am tired of all these people arriving with their theorem,” he would repeat, forgetting he had said it one, two, five years earlier. He had lost curiosity, the worst thing that can happen to someone who had made curiosity a profession.

What is more, the ambiguous impact of the technology emerging from science as a vehicle for making money disturbed him. He accepted the opportunity as a necessary

evil, with the same enthusiasm as someone starting a diet based on celery: apparently it “does good” to scientific progress.

As soon as I arrived, with their theorem, I said: “I apologize infinitely. I am immensely late.”

He did not seem annoyed and never had been. “Are your apologies a countable or an uncountable infinity? Don’t worry. If you have never missed a train, it means you have spent too much time at the station.” Wisdom he attributed to an Indian holy man, but I remember hearing it at a conference.

“Rightly so,” I answered. There followed two sincere smiles, a ream of pages, and a monologue on the beauty of the *Technical Lemma*. We talked for at least three hours.

That proof, I never understood it, and I am certain Albo Gilmor understood it even less. Yet he listened to me. In my mediocrity, he saw me as a person without ulterior motives. This opinion of his has always flattered me and added little pressure to our relationship. Sometimes I was pushed to do better, while sometimes I simply had to accept who I was. It is a liberation to appreciate one’s own limits.

“Something to eat?” He was having a kind of rejection crisis. He refused to understand the proof. We could count few people in the world with the experience required to finally put a mark of quality on the work. Gilmor had an emotional connection with the article. Yet in one side of his mind he was walking on the avenue. He imagined the moment when the chocolate mousse loses that foamy consistency to become cream on contact with the palate. It tastes different as a function of your state of mind. He was thinking: “One cannot determine the happiest day of our lives. Period.”

I hesitated for a moment, then asked, “Does it seem late to eat? It’s three o’clock.”

He threw me a glance, his eyes creased at the corners. “Jean-Paul Sartre wrote that three o’clock is always too late or too early for anything you want to do.”

“An erudite citation,” I noted, sniffing, then ordered a boeuf bourguignon, because lunching on a coffee, for me, it must be my cultural soul, was equivalent to a crime. I ate the little blocks of meat like a man possessed, brandishing my fork as if it were a dagger. Excellent caramelized pearl onions.

“Dario, do you remember the happiest day of your life?” Gilmor asked, expecting an exhaustive answer or at least one in a philosophical tone. The answer was, “O i-ea.” With my mouth full. Feeling sorry for my words, I made the effort to speak again.

“I live the happiest days every morning when I wake up after a night in which I slept well. I am lucky because lately I sleep well almost every night. I sleep badly when I dine late and go to bed right after. I have terrible dreams.” He had tilted his head. His misaligned gaze asked for more. He was patient in his feeling that I was a simpleton. He knew I was joking. I am an insecure person, and so I prefer to joke instead of saying something forcedly intelligent and thereby exposing myself to the potential judgments of my interlocutor.

“Seriously. I don’t know. Maybe the day when, at the age of ten, I bought a radio controlled toy car with my savings. I had a plastic bag full of small coins. I had set them aside by keeping the change from my mother’s grocery shopping.”

“What did she buy?”

“Bread, fruit, vegetables. Milk?”

He smiled. “I suggest chocolate mousse. It could make today the happiest day of our lives. Yours and mine.”

The document in my backpack dealt with the Goldbach conjecture, now the Goldbach-SeneQCa theorem. It has been one of the most famous problems in mathematics: every even integer greater than 2 can be expressed as the sum of two prime numbers. For example, $8 = 3 + 5$ and $10 = 5 + 5$, where 3 and 5 are prime numbers. A number is prime if divisible only by itself and by 1. This means you cannot break it into equal parts without obtaining a fraction. For example, 5 divided by 4 makes 1.25, divided by 3 makes 1.66 repeating, divided by 2 makes 2.5. So 5 is a prime number. It was a conjecture proposed in the far distant year 1742 by Christian Goldbach, in a letter addressed to Leonhard Euler. Two titans of mathematics. It resisted the attacks of the human brain for something like three hundred years. It surrendered to the electronic attack of SeneQCa. Thirty-six pages of proof. A nothing.

“Louis, mousse au chocolat, s’il vous plaît.”

“C’est le meilleur choix possible, professeur.”

“Without a shadow of doubt it is my favorite dessert. It is the password to all my accounts.” In that instant his eyes shone. He laughed with sincerity. “Now you know a secret and you will be able to access all my accounts.” I felt almost tender toward him.

“I will empty your bank account.”

I told him it was a somewhat weak password. He answered with another smile, telling me that cryptography made him anxious. He had always had faith in human beings and did not want to stop having it because of a field of mathematics designed to guarantee secrecy just because somebody wants to poke into our own damn business.

“Using cryptography acknowledges the failure of morality in human society. I see it as the saddest mathematical tale,” he said.

“Cryptography can serve to organize a surprise birthday party.”

We spent a good three quarters of an hour up and down the article before having a mental map of it. The juice was in four, maybe five pages. “People believe it is correct. By now this article has been sifted from top to bottom.”

He complained about the use of a new term: *grammarion*. I pointed out to him that new terms had been approved by human beings during the drafting of the manuscript; terms, however, invented by the machine, it seems.

There were new ideas in the article. New techniques. It was not an exhaustive case by case analysis. SeneQCa reasoned for real. Gilmor ordered a long coffee.

He told me that in 1976 it took one thousand two hundred hours of computation to prove the four color conjecture: the nations of any geographic map can be colored with only four colors so that any two bordering nations have different colors. It was a historic result for two reasons: first, because it was an important conjecture; second, because the proof was done with the help of a computer, and for this it raised misgivings in the scientific community.

SeneQCa’s proof provoked dismay in the broader scientific community. Not for me, or for those who were expecting these results precisely because they were familiar with SeneQCa’s technology. We all said: “It is only a matter of time.”

Gilmor did not like the article’s list of authors. He grumbled: “Too many authors.” According to Gilmor, more or less all of them should be deleted. SeneQCa should be left. It was a question of intellectual honesty. He fiddled with the napkin.

The mathematician Doron Zeilberger, starting in 1990, wrote various articles together with his computer. He had renamed it Shalosh B. Ekhad¹, where in modern

¹Shalosh B. Ekhad and Doron Zeilberger, A WZ Proof of Ramanujan’s Formula for Pi, *Geometry, Analysis and Mechanics*. January 1995, 107-108. Dedicated to Archimedes on his 2300th birthday.

Hebrew Shalosh means 3, Ekhad means 1, and B stands for 3B1, from the model of the machine, AT&T 3B1.

Machines will never replace mathematicians, but those who ignore them will inevitably be overtaken by those who use them. Unless the mathematicians who use machines begin to think in ways so different that they no longer require the title of mathematicians, and become operators of artificial reasoning or some other thing. And so we will speak of different categories. Formal designers, again. They will replace us all right. They will have to be kept an eye on, though, through other machines, in a chain letter in which the roles are still to be clarified.

First of all, machines are useful in searching the scientific literature. Immense archives contain repetitions upon repetitions. Proofs of intermediate results cooked and recooked by generations of mathematicians forced to reprove them, not from a desire for completeness, but only because unable to find them when needed. Machines pull the genuinely important information out of the archives. They will put it together, dishing out in front of us ready made plates already containing most of what is needed to close a proof. We will stop groping forward from zero.

Will it be a future where our role shrinks? A herd of ex-intelligent people, or nothing different from using a dictionary when we do a translation? The words are given by the dictionary. The logic by the translator.

“I would not be surprised if at some point SeneQCa proposed conjectures.”

“Right,” Gilmor continued. “Machines that ask questions will be more interesting than those that only answer. Pablo Picasso used to say it too.”

“And if a proof made by a machine were too difficult to be understood by brains? A problem.”

A mechanism is needed to verify its correctness. It is called formal verification. Automated reasoning. It uses logic, no? Who knows, one day the machines will produce scientific articles entirely without our help. Articles too difficult for us and therefore incomprehensible. With irony, they will improvise articles full of complex terms and with inscrutable proofs. They will watch us, amusing themselves, while we pretend to understand them, ashamed of the contrary.

And they will stage dozens of Sokal affairs, writing senseless articles while we remain dumbstruck during the reading. The scientist Alan Sokal, in 1996, published an

article² with no content at all, only a row of sentences that sound good and seem written with care. Once it was published, Sokal revealed that the piece was a hoax. The experiment helped to highlight the problems in the publication process, where even editors and expert reviewers are not able to filter empty manuscripts. Milton J. Cox did the exact same thing in 2035 with a mathematics article. The article was verified by a machine with an automated reasoning system more primitive than SeneQCa. The machine found a microscopic error that the astute Cox had slipped into the article. Cox made a small fool of himself as a contribution to the progress of science.

For this reason Gilmor had always complained about the lack of a standard for writing in mathematics. “Look, as you know, if you want to prove something difficult and useful, you have to prove many lemmas. You have to go to the bottom. In mathematics, perhaps one of the few fields of human knowledge, you can avoid faith and embrace logic.”

According to Gilmor, one must subdivide mathematical discourse into the elementary particles of the proof. Having many lemmas will perhaps be inelegant, but a sequence of lemmetti is easier to understand than a proof of a single statement running two full pages, where the devil finds room to hide among the missing details. In those two pages there is too much opportunity to commit errors. If, on the other hand, every lemma requires a proof of a few lines, we can breathe easier.

“Of course, you are right. I prefer someone to take me by the hand. Two key points: mathematical language should be standardized; formal verification is a necessity. And the verification can be done only by machines, because we humans cannot keep up with the complexity. We are not capable of it. We were not programmed for this. We are programmed more for creativity than for verification. Machines do it by definition.” In his words there was the faith that machines, one day, would solve mathematics, provided everything could be formalized. And yet, in his eyes I knew I read a different story, a story of sadness.

I made a pitiful joke. “Do you believe SeneQCa obtained this result by mistake? Do you think it is an accident? A fluke?” I asked him, not so much with curiosity as with

²Sokal, Alan D. “Transgressing the Boundaries: Toward a Transformative Hermeneutics of Quantum Gravity.” *Social Text* (1996): 217-252.

sarcasm, and as a provocation. I asked him while I chiseled the chocolate mousse with a small spoon and felt a good dose of happiness.

The question struck the tuning A for a *pippone* on randomness in algorithms and on the existence of emergent phenomena. He recounted at full speed the history of the universe from the Big Bang to abiogenesis, that is, the origin of life. He set off in fourth gear because I had touched a sore point in his scientific soul.

“In the end everything is an accident,” he concluded, creasing his face for a second. *Iipse dixit*. If he says so. “It depends on how vast the probability space is. That is, on how many options you have.”

And he almost got angry when I challenged him, involuntarily, by saying that algorithms always proceed in a way we can understand, including probabilistic algorithms, that is, the formal procedures that go at random. It is enough to keep the list of all the possibilities. “If the possibilities are too many, or if we do not know how to define the possibilities, what an algorithm does to arrive at the result is not determinable.”

In other words, one of the greatest luminaries of computer science was saying that SeneQCa had proved the Goldbach conjecture by sheer dumb-ass luck. I was not prepared for this answer. Meanwhile I could feel with my tongue that my front teeth were dirty with chocolate. A plausible interpretation, but I expected something else, the way Gilmor expected something more from his question about the happiest day of my life.

The process that occurred in obtaining the result is not important. What is important is the fiscal strictness of the verification process. It is not important that the monkey in front of the typewriter’s keyboard puts the Divine Comedy on paper by pressing keys at random. It is important to verify that it is the Divine Comedy. Gilmor content, me content. I saw him cough into the napkin.

“*It’s a coincidence. It’s an accident,*” he insisted on repeating, while he watched the people on the avenue, almost as if to deny, naively, the superiority of the machine he himself had helped to build. I know very well that this was, banally, a matter of legitimate narcissism.

“Albo, tell me: is SeneQCa Adam or Frankenstein?”

He laughed coarsely.

“It is a can vending machine capable of writing and reasoning, and if you talk with it you will see how half the things it says make no sense at all. Sometimes because of its stupidity, induced by the wrong statistics, and sometimes because of our lack of imagination. We have had dialogues lasting entire nights.”

“Who?”

“SeneQCa and I.”

We left the bistrot when dinnertime was approaching. The waiters said goodbye to him affectionately. We shook hands in the middle of the sidewalk. This was a three and a half years ago. And it was the last time I met Albo Gilmor.

On the metro, going home, I wrote a few notes on what defines a coincidence. A coincidence is the occurrence of two events at the same time, perceived as extraordinary and unexpected. It is not a coincidence that it is cold in winter: it is predictable. It would be a coincidence to go out without a coat on a winter day and discover that it is unusually warm. In the same way, thinking of a friend you had not seen in a long time and running into him by chance is perceived as a coincidence. One can believe that the thought contributed to making that friend appear, but one need only accept that coincidences are simply part of daily life. Some we notice, others pass unobserved.

For a computer, generating a complex output is the result of a long and careful process, often the fruit of meticulous human work in programming, statistical analysis, and inference. However, if the process includes elements of randomness, the probability of obtaining a specific result can seem low. In this sense, when a machine manages to solve many other problems successfully, it is no longer a matter of coincidence, but the fruit of effective design. All of us, naturally, waited with curiosity for SeneQCa’s new results, to understand what other possibilities it would disclose.

And yet, after meeting Gilmor I felt pessimistic about the future, when I should have been galvanized for progress. I felt instead a little useless, as if relegated to the margins in the attempt to understand something that went well beyond what I could grasp. It is easy to imagine a calculator able to perform computations faster. The process, after all, is clear and comprehensible. Thinking of a machine able to reason is another story altogether. A story that makes you curl up small, the way, as a child, you looked at the stars and suddenly felt in your throat how vast the universe was beyond the terrace.

Come to think of it, the blender in the kitchen solves a problem better than you do and does not make you feel small in any way. On the contrary, you use it gladly.

I asked myself whether even the four little things I had accomplished professionally up to that moment were not simple lucky combinations, weaves of encounters, of conversations, of threads stretched without a precise design. Perhaps chance, like an invisible orchestra conductor, arranged ideas in a merely apparent order, juxtaposing them, aligning them, leaving me the illusion of having chosen them. Perhaps even those mediocre theorems I had proved were nothing but the result of being in the right place at the right moment, of having read the right article on one particular winter morning, of having met that particular coauthor at a conference. I accepted calculators, and I believed that reasoning was a defining process of our humanity.

The concept of something highly improbable brought into my thoughts the memory of an event from many years earlier. I had unplugged the radio alarm clock. During the night, the green light of the little numbers bothered me. It created a halo on the other side of the room, on the wall, across the small bookcase and onto the two doors of the wardrobe. A useless half moon. A distraction even while you sleep. Sometimes I moved the clock in the middle of the night. I had set it with the luminous face turned against the wall. This was not a good solution because, although it no longer lit the whole room, it planted a greenish parabola, dragging itself up from the nightstand to three quarters of the wall. Sometimes I put it directly on the floor, on the other side of the nightstand, stretching out like a goalkeeper diving to save a penalty.

The little light was a fixed thought, and it came back to my head every evening when I switched off the lamp. If you put together those two seconds in which I thought about the little light, you see that the total time becomes hours, adding up the two seconds over days and days. Was I supposed to throw away hours of my life, whose time is limited, thinking about the numbers of a radio alarm clock? An action was needed to close off space for this thought. One night, in fact, I opted for the most drastic solution, and perhaps the best one, considering that I never used the radio alarm: unplug it. So I did. I pulled the plug from the wall and instantly the green numbers disappeared from the face. Maybe I passed that night better than usual. I don't remember well.

Some time later, I am talking weeks or months, in a late afternoon, I decided to plug it back in. Again, not because I needed the radio alarm, but for some reason connected

to the order of things in my room and to the unconscious desire to see whether it still worked. I expected to find blinking zeros. Instead it was not so.

The clock read an anonymous 6:57, probably not far from the actual time. I looked at my wristwatch. It was a useless gesture, because I persisted in keeping on my wrist a watch that had been broken for I don't know how long. I should have thrown it away, set it aside in a drawer, or gotten a new one. Repairing it made no sense.

It was worth between little and nothing even from the affective point of view. Not keeping correct pace, it sometimes marked hours very far from the real hour. I kept it out of habit.

For a moment I paid no attention. I looked at the wristwatch, again, for more than a second this time. On the small face I read, without error, 6:56. I went into the other room, the kitchen. There was another clock, and it worked. A wall clock. I read 6:59. Three clocks: one long unplugged and switched back on at random; one random; one hanging on the wall and working. Let us say the same hour, 6:5x, taking an average.

I calculated the probability of the coincidence while I ate dinner, just to give further rope to the mathematical obsession. I threw the wristwatch into a bin and returned to my nightly struggle with the radio alarm clock. I still have it, shut in a box somewhere. It is an irrelevant object, like those theorems of my scientific career, on which I wasted an absurd volume of time, as if they were colossal and useless sudokus.

Tuesday, December 12, early afternoon

Two people of indeterminate features are running after me. I climb the stairs. I trudge. Ten steps. I try to escape them. The more I climb, the more my flight slows. I tremble and try to wake up. I cannot. They are two school janitors. The mathematics exam has begun.

I open my eyes, sitting up sharply at the exact center of the couch. This afternoon the light filters through the blinds and projects horizontal shadows into the room. They are barely sketched shadows, thanks to the light of Seattle in December, which knows how to be decidedly depressing. Here they say "the mountain is out" when the sky is clear and Mount Rainier can be seen from the city. Today it is cloudy. I look at the green

blanket. It reaches from my waist to my feet. The fabric is frayed. Familiar. I remember my first mathematics exam at university. Almost everyone copied, as in a chain reaction, starting from the two best girls, who never copied but served as the source of the information; or they cheated, many of them, with graphing calculators that could plot functions and give you a sense of the analysis problem.

I arrived late, and for this I had to sit one meter from the professor. I did not copy and I had no instruments to help me. In those days I was unfocused, inattentive. I had not understood that what was needed was reasoning more than memory. Had I realized it earlier, maybe I would have developed a different method of study. Failed, according to justice. I tried again a few months later. I passed with the minimum grade. The irrefutable proof of my scarce talent.

Today, I have been without a job for some time now. From the financial point of view I am calm. The compensation obtained for exiting the startup should last me a long while, provided I maintain this lifestyle and commit no imprudences. The initial package of shares they took back, in part, through various subterfuges. I was fired. Technically, at least.

What to do with my abundant time? Set myself to doing something decent in mathematics? Absolutely not. I have stopped, aware that there is no going back. Speaking frankly, I have never been good at mathematics. Maybe in middle school, when the numbers were still kind. Then something changed. I rarely find concentration. I tire immediately. I get lost. I am easily distracted, even in front of simple exercises. It is as if my mind slid away at the first sign of abstraction. And with the years, instead of improving, I have gotten worse. Even in the most elementary things I seem to glimpse traps, ambiguities hidden among the symbols. The numbers tell me nothing. They are hieroglyphs of a civilization forgotten and far from mine.

But I am curious. So in an inner concavity, in a recess of the soul, I perceive their beauty. I have no doubts. It is a beauty incomprehensible to me. But do comprehensible forms of beauty exist? Beauty is felt and that is all. It is not understood and should not be understood, despite attempts, rather strange ones, to define its characteristics, sometimes even from the numerical side. Today I am at my minimum in the capacity for concentration. I have dancing thoughts, disconnected from one another, like leaves pushed by the wind in autumn.

And yet, in certain respects, I have mathematized my life as much as possible. I do not know whether I did it out of a strong attachment to the maternal figure, out of an instinctive desire for harmony, or perhaps because I have always suffered from a chronic condition of adolescent maximalism, the kind that in my case pushes me to give a meaning to everything, even when there is no meaning and no need to look for one. I refer to a continuous search for reason and to being utterly averse to those risks owed to a lack of information.

From a psychological point of view, if a mathematical therapist existed, I would without doubt be a difficult case. He would need much patience and creativity to help me rebuild my relationship with mathematics, and maybe, in the meantime, with myself as well, starting completely from zero.

This thought on the mathematization of life passes tangent to my head. It grazes my hair. While I toss into my mouth two greasy potato croquettes taken from a paper bag on the coffee table, I hear a beep from the phone. To avoid smearing it, I grip it with a digital contortionist's move and a dangerous game of pinkies.

I read: *the struggle of maturity is to recover the seriousness of a child at play, friedrich nietzsche*. All lowercase. Nietzsche. No less.

An unwanted advertising message? A harasser? The message was sent from my own number. Meaning I sent it to myself? Then it is an automatic message sent by a computer. Of course. This quotation, what does it intend? Now, of all times. Am I perhaps not a serious man?

The problem is not machines more intelligent than Homo sapiens. The problem is constituted by the Homo sapiens who are bigger assholes than the machines. If we do not like the machines, then we can simply not use them. It is we who program them, these machines. They say: "They are dangerous when their behavior slips out of our hands." Sure, when you slice bread with a knife you have to be careful. But what if these machines at a certain point are agents capable of making decisions we do not understand, or decisions against our interest?

Clearly, they will do it better than we do, and they will take this responsibility away from us. It is an idea at once frightening and liberating. If the machines took control, maybe they would finally manage to untangle the complex web of human greed and incompetence. The weight of decisions, with all its potential for catastrophic errors,

would pass from us to them. We would no longer be the architects of our own ruin, but rather passengers on a cruise ship.

You cannot tie all the grass into one bundle. In what way would machines that prove theorems work to our detriment? A not so unique reflection of our stupidity, or with strange insular interests, could they prove useless things and thus pollute the sea of mathematical knowledge? We, at the center of the universe, interpret them as rivals, these contemporary machines. As antagonists. As something that could induce us to lose the meaning of life. I do not believe it. I chew another potato croquette. It has a smell different from its taste. I think of Aesop: the hare and the tortoise, the fox and the grapes. The anthropomorphization of animals. Here we go again. It is all a fine narrative we tell ourselves to give meaning to a changing world.

A few minutes after receiving the message on my cell phone, a call arrives from an unknown number. The code is 207. I look it up in record time even before answering.

I ask the phone: "Where is code 207 from?"

The synthetic voice answers, decisive: "Do you want to know how to cook cod?"

Excellent. Getting from "code" to "cod" takes creativity. Let us do it by hand. Maine. Who lives in Maine? I answer anyway, perhaps out of boredom.

On the other end is TH. I should have imagined. After predictable pleasantries about various microclimates, about the treacherous ice and the untrustworthy road surface, in this extraordinarily cold December, he finally gets to the point:

"You have to ask me for the password."

Right, I had forgotten. "What's the password?"

"Hairy spiders twentyfive." A password by now beyond embarrassment.

"Thank you, Dario. It seems stupid, and yet it is useful."

"It does not seem stupid. It is stupid."

Albo Gilmor has had an incident, and TH does not know whether to consider it an incident. He prefers not to explain the details to me by phone. He fears malevolent ears listening. He alludes to difficult legal questions, to my connection with Gilmor and to the way I could be of help to him. And at this point, the sense escapes me entirely: it is as if the words, taken one by one, said something, but once placed side by side, managed only to confuse, decreasing rather than increasing the informational content.

He asks me, "Would you talk with Sandra face to face, please?"

“All right,” I answer. And I feel how much I like answering “all right.” *Sure*. It removes the weight of contradiction and spares me the necessary expense of energy to sustain one. It is a small comfort in an uncertain world.

Gilmor is no youngster, and TH does not tell me where and how the incident happened. Whether in a car, on foot, in the shower, whether he fell down the stairs or cut himself with a razor blade, or scalded himself while cooking two fried eggs, or whether he had a health problem, whether his heart stopped, or his head, whether he fainted, whether he took the wrong medicine, or forgot to take it, or whether his fingers got trapped while he was tucking the sheet under the mattress, a terrible thing, and one of those domestic situations that make no news but leave a mark. Sharp pain, surprise, a sense of powerlessness. The kind of incident that makes you reevaluate the design of the modern bed, by now taken for granted. Statistically it is more probable that he fell at home, or a sudden illness. TH is sibylline. My mind runs fast toward various conclusions, but none is worrying.

“Is he all right?” I ask, with reason.

“You should meet Sandra Bianchi. Talk to her. She has something to give you.”

“Excellent,” I re-reply, and I say: “I have just received a message sent from my own number.”

“We know. Answer it if you want, let us see what happens. Actually, stop. Pay it no mind. We will know more about these messages in the coming days. You are not the only one receiving strange messages.”

Sandra is at a conference on the use of high performance computing. They talk about how to simulate physics. In the past I too have been to that same conference. Various editions, maybe five or six. There is a plane soon, at a decent hour. I take the cheapest possible ticket through an automated agent. The familiar routine of conference travel crosses my mind: the crowded halls, the late night discussions, the enormous quantity of chatter and gossip.

“Please, do not talk about this with anyone. Courage.” TH closes the call.

The news of Albo Gilmor’s incident must remain confidential. I look for it on the Internet and do not find it. It must be a minor thing, and I hope it is. I have a bad presentiment, though. Usually, when TH talks, he alternates sentences and snickers. This time he seemed serious to me. I pile the blanket on the couch. I take my backpack.

Into it I throw a toothbrush, half a tube of toothpaste, a pair of underwear, a T-shirt. Socks. I clump two cables together and slip them into a pocket. The laptop goes in the space at the back. Vertical. I look at the backpack. It is half empty.

This is why I am an optimist. A half empty backpack gives me a way to add other things. Not like a half empty glass, where I am interested in the total quantity of liquid I can drink. Packing the backpack calms me. I go.

The phone rings again. It is TH. I answer. "Hairy spiders twentyfive."

"Gilmor did not make it."

"Meaning?"

"He died in the incident. On the spot."

I have not seen Gilmor in several years, but dead means dead.

"Excuse me, TH. We spoke thirty seconds ago. Why did you not tell me immediately?"

"I thought it was not an important detail. I changed my mind. Sorry."

"And what is the important detail?"

"The relevant point is the incident, not the effect of the incident. It is the cause of the incident."

"Are you serious? Has his daughter, Angela, been told?"

"Yes. Help us understand how it happened. We are tangled in tedious legal matters. Never mind. Gilmor left a note with your number, there must be a reason."

"The thing is a tiny bit unsettling. Where did he leave it?"

"At his house. In the kitchen, fixed to the refrigerator with a souvenir from Amsterdam."

"A what?"

"A magnet. A windmill..."

I hang up in his face. I take in the bad news with the necessary resignation and a light sadness, hard to process. It is a sudden death that happened inside an obvious window of time, given Gilmor's age. I replay memories at random. Gilmor sitting with his legs crossed and a glass in his hand. He said: "Try to do something particularly human."

"For example?"

"Have fun."

Lately there is nothing but talk of the dangers of artificial intelligence, of sentient machines ready to supplant human beings. And yet this public debate seems to me ever more vague and superficial. It is a constant buzz, devoid of clear definitions of what intelligence really is, or of what these apocalyptic scenarios that so fascinate us actually mean. The common discourse does not try to inform, only to attract attention, exploiting the same worn commonplaces taken from decades of science fiction.

And so, in reaction, I take refuge in an old mental game: what if, instead of looking at machines as human beings, we tried looking at human beings as machines? Would that not be a more productive exercise and a way to restore order? Maybe. Mechanicism diminishes us, it lacks magic, and what is more, we human beings are control maniacs precisely because of our fear of the unpredictable. We want to dominate it, not deny it. So we want free will.

In this obsession of ours with control, we end up forgetting that maybe our consciousness is a simple monitor, an overseer of actions and decisions. It is a sad vision, I admit, even if evolutionarily good for survival. I also know that, by forcing ourselves to set the magical aside, we leave out something essential. Because all of nature is deeply magical, or at least it becomes so the moment we try to recount it with our stories. And science, in the end, is nothing but our most stubborn and rigorous attempt to give a meaning to that magic, to translate it.

It is a principle of every aspect of existence: from the birth of a child to photosynthesis, all the way to the formation of stars. Or maybe I confuse poetry and magic which, rhyme aside, are perhaps two different ways of realizing the same, identical idea.

So, while we optimize life, we feel a touching warmth in thinking that we are special anyway. And what if we always ran on autopilot? And what if free will were only an illusion? And what if we were merely expert assigners of meaning? A meaning emerging from the contact between the world and our senses. That is how we differ from the machines. We give personal meanings to everything. Maybe we are just labeling machines, we human beings, because meaning is a necessity for going forward, for giving ourselves a motivation. Can one really live without meaning? Then comes creativity. Machines only put together things already seen under the sun. Do we do something special? I doubt it.

Honestly, it does not interest me. It is a matter for experts. I would just like to let myself go. To do what I feel like doing. For example, I would like to write a few short stories. Not autobiographical. I would like to write a story with imaginary characters, just to begin. I would like to do it because it amuses me, without asking myself too many questions. I am sure I will find the time and the willpower to write, sooner or later, a silly and senseless book. It will be an essay on an absolutely idiotic subject. Maybe I will write a story about an explorer lost in the shadowy hollows of a remote forest. A tale in which the imagination is free to wander through treacherous lands, where, safe among the pages, beautiful, white, clean, everything is perfectly harmless, as a metaphor of our invented fears. No drama for me: no malaria, no risk of getting lost and never returning, and not even the extremely dangerous necessity of making decisions, which by the way is also a privilege, whose denial is a form of prison. All of this unlike the real world, where a moment is enough to lose your way. It is enough to get distracted, to not be present, to not know what to do next.

The only purpose of my story will be to laugh while I tell myself absurdities. The explorer's journey could become a metaphor for our effort toward technology or discovery. On one hand, it would show how hard we work to answer the great questions. On the other, it would reveal our laughable tendency to overvalue what we do, convinced of a deep meaning and epochal consequences, even when of meaning there is barely a shadow. An inexplicably physiological narcissism. I will write a story on the vain side of ambitions. A reminder: however hard we try, certain things will remain outside our control. Or, more simply, they are not destined to be understood, returning to magic. But then what would the alternative be to this vain explorer? Maybe another type of traveler, one who does not try to conquer the world but to recount it, and to try to complete puzzles of little value, with the same dedication as a child absorbed in his game, like the message just received. And here is the idea of seriousness. Being serious, maybe, means working with your mouth closed. With dignity and above all believing in it. As in a formal system represented in life, where we carry out tangible tasks and, precisely for this, feel we are in our place, sometimes translating seriousness into the comic.

About the message on the phone. Was this what Gilmor meant, speaking of something particularly human? Facing life with the commitment of a child? Believing in

it. Better not to speculate stupidly. In play we lose the awareness of having a life outside the play. We forget it. The problem concerns the sustainability of our attitude in the game and its compatibility with personal wellbeing.

And yet in play we have a specific and authentic task, an easy thing to neglect in life, where it is simple to convince oneself of the value of useless things.

I leave the house with the weight of the legacy left by Gilmor, great scientist, but also a man whose private life seemed almost not to exist, so hidden was it and pushed into the background. Climbing into the taxi, I surprise myself thinking that maybe the true reason for this trip is not meeting Sandra to do TH a favor, with his dynamics and his odd bureaucracy. It is to pay a tribute. A way to draw close to Albo again. Maybe the particularly human thing to do, as he would have said, today as always, is exactly this: to face every mystery, every assignment, with an open mind and seriousness. And maybe even manage to find a little humor in the process, however absurd it may seem. To stay adherent to the contingent, to the immediate, without extrapolating grand theories. I close the taxi door, careful not to slam it, while the auto-driver takes the road for the airport. In this jumble of thoughts, physical reality brings me back to earth: with my tongue, I remove a residue of croquette stuck between my teeth.

Tuesday, December 12, late afternoon

I am on the plane. When I go many hours without eating, my thoughts become sharpened. Brief. Irritation rises over nothing, induced also by my psychological Carnot engine. That is, I feed it myself. Everything, though, loses thickness, as if seen from far away. Hunger is no longer a need, it is a filter: everything arrives distorted, emptied, and ready to begin an introspective domino of mental states. And in that emptiness of the stomach, but swelling of the brain, there is also a vague sense of lightness, as if the body stopped asking and left the mind wobbling and free. There, in that going of hers, the mind falls into cavities, like steep spirals, and I, inert, stand watching her, with a little suffering. I would be tempted to say like a boxer on the ropes, hit in the ribs. It is not a good metaphor. I watch her as a spectator, exactly, as now, returning to the message received today.

Good for Nietzsche, scattering his aphorisms like chicken feed. What does the sender want to tell me? It could mean that I am wasting time, that I am playing without really believing in it. Or maybe, on the contrary, that I take everything too seriously. It could even be a threat, however elegant, or an encouragement to continue, a signal that I am getting close to something. To continue what? Maybe it is an invitation to begin the real game again, the one that for me has always been mathematics. In the end, though, the simplest hypothesis is also the most bitter: maybe it is just a way of reminding me that I am getting old, and that the time for playing is running out. And if I do not seize it now, I never will, and like Gilmore I will end up as food for earthworms. What is the truth?

Certainties are risky. Certainties are anthropologically useful. There are facts and there are statements. Sentences about the physical world or the abstract one. True or false. Or undecidable, as formal logic teaches with its celebrated incompleteness theorems. To be certain about human things is not, I will not say hopeless, but almost. A formal system is needed. And in 999% of cases, none is available.

So we wander in circles, no longer even claiming to understand whether reality is the narrative we are weakly convinced of, or whether reality really forms a stable construct, where someone, with more capacity for logical thought or more information than we have, could tell us what to do and what not to do. Maybe even make it comprehensible and explain it to us in part, or at least tell us some semblance of truth, in case they did not want to share some secret with us. A superhuman entity, custodian of the truth.

I have been urged so much to push toward the search for truth. "*Are you a truth seeker?*" I ask myself, and they ask me. They do it almost in a tone of challenge. For what? The search is necessary. To feel well? Yes, to live without hiding the dust under the carpets. To not stay fossilized in our beliefs. Maybe to feel better? Maybe as a necessary trait of the pilgrimage in search of ourselves? To find harmony. I am sorry about Gilmore's death.

One must seek the true in order to distinguish, to the maximum, that is, with the maximum extension, imagination from the real. Mixing imagination into the real is dangerous. Period. Unless it is declared openly. In that case we speak of theater.

One needs to compare notes with other people to build this reality, to certify it. One must take averages. Set aside the convictions generated in nocturnal solitude. Others serve to regulate us. So reality is built in a group. Unless it is the mathematical reality built in the laboratory and in the safety of formal systems.

To ascertain human, non-mathematical reality, then, a great orgy of communication is needed. The secluded onanism of the philosophers' thought produces only a personal interpretation. Shareable, perhaps convincing, but personal. Like a solo crossing, where in the night islands alternate with mirages. A risk. Thinking without comparing notes with others generates risks. To determine whether what I see is a hallucination, I must ask those seated next to me. Maybe this is the true, deep difference between the machines built today and people: people are big containers of neurons that bump into one another, with a little randomness, but not even too much. They enrich themselves with their senses, they modify themselves, they share, and they regulate one another. People are surprised by epiphanies when they meet. They have very many sensors for gathering signals from the external world and interpreting them according to the information they already possess. The machines stand still, insensitive, while we spoon-feed them, and they spit out the result of products of large matrices, almost with the intent of humiliating us without making a sound.

If one wants, one can exchange opinions with a preferred interlocutor. The one-to-one linguistic connection helps. At bottom, though, one must compare notes at length with many other people. A group conversation is truly needed at the moment logic is missing. The whole of reality is not formalizable, and maybe it never will be.

But what is cultural and what is human? Do you accept the truth once learned? Do you pretend not to hear it? And how long you drag it out, though. You invent a parallel story out of fear or out of laziness. Or rather, the one inside the other, seeing how laziness is perhaps a phenotype of fear. There is, however, another angle. If the reality to lean on really requires an average among many personal realities, then only a machine with the task of weighing all these realities and putting them together, and measuring their differences, could really help us determine what defines us. The machine, taking the average, vaguely speaking, would understand what common sense is, observing it through a statistical analysis. So a machine is needed to understand the characteristics of human beings. Characteristics in part listed by us, and in part generated by the

machine itself, and therefore impenetrable to us. It will be the machines that teach us anthropology. Paradoxical. I am sorry about Gilmore's death.

Careful, there are also facts. Measurable, determinable. A fact is the plane being late. I do not know when we arrive. Nobody has told us. I have purchased Internet access from the airplane seat. I do it with a light sense of guilt. To send half an extra email, to look at a few superfluous messages on the phone. My messages. Nobody has written to me. Moreover I know well that nobody will write to me. Maybe Friedrich Nietzsche will write to me again. Let us see. I am sorry about Gilmore.

This is the shabby reality of our addiction to electronic instruments. Technology useful only at times. We are deceived through loopholes of evolution. The instruments do not succeed on the basis of having found niches in the markets. They find holes in our psychology. And they filled them at once. Like that microscopic pleasure of being always in contact. Pushing the images upward on the little screen. With the same finger. Self gratification. Rituals. What anguish. Among other things I feel the beginnings of tendinitis in my thumb.

In 1909, with the story *The Machine Stops*³, E. M. Forster described a world totally dependent on a machine, a machine so central in the life of all people as to be elevated to a divinity. People live underground, in single little rooms suited to satisfying the needs of the body but distant from one another. For this reason they communicate through a system for exchanging messages at a distance.

Does the situation seem vaguely familiar? Once Foster's machine stops, civil society ends up stopping too. A brutal resemblance to our deep dependence on machines. Deep to the point of making me look at the phone every five, at most ten minutes. Deep to the point of having helped turn me into a physiological superficialist. To read a page of a book I need an infinite amount of time. After two lines I am lost. I am forced to look at the phone. After another two lines my mind already requires a homeopathic dose of another type of information. Going beyond titles and subtitles demands a Herculean effort.

I look out the window and my mind returns to me. Landing is not far off. I am giving in to bad humor. I use this fragment of time to look at where we are on the map while I

³Modern Short Stories, S. H. Burton ed., Longman Heritage of Literature series, Longman Group Ltd, Great Britain, 1965, sixth impression 1970.

drink a chamomile from a paper cup. The plane has taken the wide way around. We pass over Loma Chiquita. We arrive at San Jose at 8:57 pm. We are 25 minutes late.

I get off among the last, though I could have been among the first. I was middle. The last shall be first, but where will the first end up? They will be second, right behind the first. Will they be last? Or will they end up in the middle of the crush? With the middles. The group without infamy and without praise. There, from a middle, I end up among the last. Some of the last have taken my place, in the vain attempt to substitute themselves for the first.

Every simple moral principle is, at bottom, a technological instrument. Like “the last shall be first,” a social technology designed to give you hope and push when life seems to row against you. These narratives as mechanisms for guiding our actions. Culture is a form of technology. Hard to perceive it as such because it wraps us constantly. And yet it is an intricate system of beliefs, norms, and tales meant to shape our perception of the world. It is an invisible technology of immense power. And one could say the same thing, but flipping the perspective: technology is a product of culture in its specific and distinctive form. So then a simple maxim like “the last shall be first” becomes a crucial gear. It whispers to you that even if today you are at the back of the line, tomorrow you could find yourself at the front. A fine illusion. Like a rigged scratch card: it makes you hope while it screws you.

Thinking it over, in society we give importance to the kurtosis of the distribution. If you imagine a bell curve of the distribution, kurtosis tells you how tall and narrow, or low and wide, that bell is at the center. The good are trained to become excellent, to make them leap out, as “champions,” a term whose applicability seems strange to me, seeing how the great part of all of us is in the center, toward the bell of the distribution. If we took a person at random, doing precisely a sampling, would we end up taking someone toward the center of the distribution? I am among the middles. “Hello, middle, how are you?” Distributional normality. Instead of improving the many at the center, the already optimized in the right tail of the distribution, constituted by the champions, get improved. Will the champions, as such, manage to accomplish marvelous deeds for the rest of us all? Will the superheroes be the ones to push society toward better shores? Will they manage, with their rare capacities, to lift all of us middles one step higher on

our ladder of mediocrity? Maybe society's biggest bet is finding a way to improve us middles. Or maybe not, because we do not deserve it.

I exit the plane through the boarding sleeve. I do not remember the seat I was sitting in. I do not remember whether I ate anything today, setting aside the horrendous potato croquettes. I do not remember the address of the hotel. Certainty is missing. I realize I am about to face whatever Sandra Bianchi has to tell me about Gilmor. I am sorry about Gilmor.

Crossing the airport, my mind jumps between rambling thoughts on the nature of reality and the unstimulating task of finding the right belt for baggage claim. It is not a problem. I have only my carry-on. What a beauty. It is the small things that make you feel good. I grip the phone and set off in the direction indicated by the sign Ground Transportation. I open the ridesharing service and I have one certainty: someone will let me into a car within a few minutes. If there is no one, then an autonomous vehicle will arrive with the auto-driver. My certainty is given by previous experience and by the fact that today there are no *unusual situations*. It is a day like any other. It is middle. My certainty has been built, precisely, little by little through the observation of past events. Experiments. The particularity of an experiment is that the result is always the same if the conditions are left identical. No small lesson. Today the conditions seem to me very similar to so many past times. The value of a middle thing is its predictability. I do not have certainty, though. I have a belief, based on empirical observations. I wonder whether one day science will no longer look for closed formulas but only for a basin, an immense reservoir of empirical facts to be used for training machines.

I see the license plate from afar. It is my ride. I get in and go. There are two possible lives, maybe among the many. The interior life, the one spent in dialogue with a part of ourselves destined for reasoning. It is a part that, paradoxically, almost does not belong to us. We observe it, we interact with it, we try to bend it to our will, as one would with muscles during a training session of weightlifting or running. It is an effort. This is the life of the mathematician, who lives in that interior dialogue, sometimes with joy, or, as in my case, so many times, with difficulty. I am sorry about Gilmor. Of course, this life is never completely isolated. One must interact with others to have more strength, to prove together what alone seems inaccessible. It is a strange sharing, of a technical nature, where one must not trust the person, only certain notions or capacities: we exchange

mental resources, small alter egos that contribute to the reasoning, these fragments not entirely ours. It is an experience of collaboration that, though it happens between people, remains rooted in the solitude of thought. It is not a sharing, it is a contributing together, with private instruments and spendable qualities, to the solution of a problem that afflicts us in common. The only thing we share is an underground torment over the conjecture not yet proved, curable only by the proof or by oblivion. We can collaborate because we have automatic formal verification.

Then there is another life, among the possible ones, the one guided by authentic interaction, by the perception of the other not as an instrument but as an entity to contemplate, and a different sharing, I do not know whether deeper, where the aim is not to arrive at a proof, but simply to perceive the existence of others, to welcome it, to let it be, without a functional dynamic at the base of the relationship. This second way of living requires openness, a type of sensitivity often missing in those inside the bubble of the interior life, even if only for “professional” reasons.

That song by the Alan Parsons Project⁴ comes to mind: [quoted lines retained from the original]. The interior life can push toward a continuous race, in which we lose the point of contact with the people we love. One becomes incapable of seeing the world through the eyes of those one meets: they are not, once again, functional to the solution of our mathematical problem. Sad? Exaggerated? Let us just call it opportunistic realism. Paradoxically, those who do science should have the cognitive instruments to overcome this myopia. They should be capable of contemplating the other. Or maybe it is simply the slavery of ambition, a common thing, sadder and more fetid.

I would like to know whether Gilmor had the lucidity to step off the race, at least once in a while. Of his private life I knew little, and he himself never spoke of it; I do not know whether out of reserve, out of disinterest, or maybe because he was afraid of it. I never met his wife, but I remember his daughter, Angela. She must be some fifteen years younger than I am. The only clear memory I have of her goes back to when her father brought her along during a study week in Cambridge. One evening we cooked together a dish of spaghetti with little tomatoes of various colors; we were in the company of other people. After dinner we played cards, a version of rummy, I believe, though the exact

⁴The Alan Parsons Project, Pyramid (1978), “The Traveller”.

rules I no longer remember. It was then that Gilmor, with that propensity of his for tracing physical complexity in every daily gesture, illustrated for us the refined grammar necessary for shuffling a deck of cards. It is a ritual aimed at deliberately sabotaging order and creating a sequence finally free of ties to the past. “A mixture of numbers and illusions,” he said with a smile.

Angela still wore braces. Halfway through the evening she got up, headphones on her ears, a quick “ciao” to everyone, then disappeared into her room. Probably a video game. She wore a black T-shirt with a faded skull. She did not talk much. She did not seem at war with her father, she simply lived elsewhere, in a teenager’s world. I remember that the next day she left early, to go back to school.

Tuesday, December 12, evening

In the taxi from the airport I read a news bulletin for the 15th time. The vote will be Saturday. Small protest marches, scattered across the country and with rather repetitive media coverage. It is right that I now provide a snapshot of the current picture. As we well know, the government cabinet is composed of eleven people from the technical group. The President of the State, plus ten members: five Conservatives and five Progressives. On Saturday, the cabinet will vote confidence or no confidence in the President. If the result is confidence, then the President stays in office. No confidence means early elections and, it may be, a change of President. A situation seen countless times in the past. At least seven votes of this kind can be counted in the last twenty years.

Since in every one of those votes it took weeks to reach a majority of five plus one, and in the last one there was a total stalemate of over seven months, with complicated repercussions for the nation, to avoid history repeating itself the constitution was changed in September and a super partes member was inserted into the cabinet. So today the cabinet is composed of twelve members. One of them, however, is a machine, and as it happens it is called SeneQC_a AA, where “AA” stands for *Algorithmic Advisor*. This time I did not help choose the name. It recalls the famous and terrifying HAL

9000⁵, Heuristically Programmed Algorithmic Computer, from the old Kubrick film. The original name of SeneQCa AA was Computational Helper. Since it made you think of a multipurpose kitchen robot, AA won out.

The computer took its name from our startup. It was the only thing we built. The only product, if we want to call it that. In the end, our startup and the computer became the same thing in the eyes of people. I will use he and she, referring to SeneQCa as the computer and as the machine, respectively, and without distinction.

SeneQCa does calculations. He ingests data continuously. Or rather, not continuously, every 1.2 minutes to be exact. He digests it in his belly. Today, according to the company's abundant press releases, the machine solves problems of various kinds, even though she was conceived to do proofs in mathematics, so much so that over 97% of the scientific articles in mathematics had been loaded into her memory, including specialist journals and archives containing publications. It was no small job.

I was among the “emergentists”: it will be enough to give all this data to SeneQCa and some special behavior will emerge, after a phase transition. I would be curious to know how far this idea has come true. Unfortunately, no longer working at the startup, I do not have the opportunity. Maybe I should call Melvin Chung. I was fired partly through his fault, so, no thank you.

SeneQCa is composed of three parts. The first part, which is also the physically bulkiest, is a classical supercomputer that has reached a beastly Rmax of I do not know how many exaFLOPS. It uses certain chips as big as piadine romagnole. They all talk to one another with extremely low latencies. The second part is a quantum processor with a million physical qubits and I do not know how many logical qubits. This subject was explained to me by various friends who work in the field, and to this day I do not follow it well for lack of knowledge. The third part is constituted by the memory devices. They are distributed on the cloud, and do not ask me where they are physically. I do not know. Maybe even in Earth orbit.

Some time back the wagging tongues said, “There is no quantum computer inside, and if there is, it produces white noise, because it is noisier than an out-of-tune piano.” Recently they have stopped saying it, given the great number of tests performed, among

⁵Clarke, Arthur C. 2001: *A Space Odyssey*. New York: ACE, 2018. See also Alfred, Randy (January 12, 2011). “HAL of a Computer”. *Wired*.

them the solution of problems known not to be efficiently solvable, given their computational complexity, except with a genuinely quantum processor. These are legitimate experiments based on logic circuits for solving real problems rather than mathematical artifacts. The quantum part of SeneQCa is used almost entirely to do linear algebra calculations. Whoever, with poetic venality, believes the quantum computer gives SeneQCa self-awareness is entirely free to do so. I think SeneQCa has as much self-awareness as my left shoe.

The only way to interact with SeneQCa is through a terminal sited in a SCIF, that is, a *sensitive compartmented information facility*. In other words, in a room to which only a very few people have access, isolated from electrostatic fields. I find this detail fairly ironic. Almost like a form of communication in a single direction, SeneQCa sucks in data and squeezes reports written by human beings specifically for his palate. Yet it is difficult for us to consult her, and the machine exchanges two friendly words only with some high priest who can enter the Sanctum Sanctorum. An oracle in continuous listening who every now and then deigns to have his say. I would call it a funnel more than an oracle.

The choice of the SCIF was made to avoid damage potentially caused by dangerous state actors or simple malefactors. It was the result of extremely long discussions, during which more or less everyone had their say, including moral philosophers, philosophical moralists, philosophical computer scientists, computer scientist philosophers, and then the builders, politicians, pollsters, industrialists, advisors, administrators, diplomats, security experts, analysts of every type, economists, systems engineers, computational sociologists, techno-optimists, bioethicists by the token, scenario predictors, venture capitalists, digital evangelists, corporate storytellers, content curators, technocrats, process facilitators, invisible stakeholders, policy influencers, innovation managers, semiologists of complexity, ecosystem architects, trust engineers, theorists of change, discontinuity specialists, and supporters of open source.

And also a certain Jurgen who came in person to all the meetings, wearing an extremely well ironed white shirt, and always sat in the same place, in the corner near the window. I believed he was one of the fab team, that is, the chip fabrication plant. No. He was the founder of some association for the rights of computers. He stopped coming

when a new type of chip made of biological material was launched. It was more interesting. The only biological thing SeneQCa had was a few gnats deceased from exhaustion on the metal box around the processor.

The input for the machine can be spoken language, though still articulated with great precision. And every time I speak of it, I cannot avoid imagining Dario Fo performing a *grammelot*: a speech devoid of sense, yet incredibly eloquent, exactly as a pompous super technician lost in his incomprehensible details would do.

The output is certainly not 42, as in the answer to the ultimate question about life, the universe, and everything, put to the grandiose Deep Thought of Douglas Adams⁶. The output is composed of two parts: the result and the annotations. The result of the question “What will next year’s unemployment rate be?” will correspond to a certain probability. The annotations are nothing but explanations of how the machine arrived at the result. Sometimes they are there and sometimes not. Verification is needed, and the annotations are useful for verification when it cannot be performed with formal methods.

SeneQCa’s quantum processor is built with a technology that uses qubits in a cryogenic environment, but which I do not know how to explain. In broad strokes it is superconductive. The connections between the individual qubits are unknown and there is no way to make a map of them. They evolve constantly, a little as in the human brain, where the neurons communicate with one another through the synapses, and the network of connections varies over time. Experts have explained to me that these connections are determined by the history of the machine, by the problems solved, by new data, and by the input for each problem to be solved.

The dynamics of the interaction seem to have led SeneQCa to be a good advisor on questions of a political character, and not only. In fact, he has produced fairly accurate predictions. He deserves credit, poor thing. Nevertheless it is a black box, or better, a gray one. In other words, today we do not know the internal workings of the computer. We can try to understand it after the fact through the annotations, which is why it is good to have not only the result as output. It is not clear, then, which algorithm SeneQCa chooses on the basis of the problem fed in as input. So even those who interact

⁶Adams, Douglas. 2007. *The Hitchhiker’s Guide to the Galaxy*. Hitchhiker’s Guide to the Galaxy 1. New York, NY: Random House.

with the machine have no visibility into what SeneQCa does, who, without getting too flustered, might even make a phone call to a friend of his, an electronic cash register serving in a pizzeria in Patagonia, and ask him for the solution to the problem. Legend has it that a developer of SeneQCa's software decided one night to ask the machine whether to divorce her husband or not. It seems SeneQCa answered "Yes," and within a few months the developer became the happiest woman in the world, once she went ahead with the divorce. It was later verified that SeneQCa had not understood the question.

In the now distant 2022, Michael Jordan, a noted expert in machine learning, said that when you develop artificial intelligence in systems like search engines, if you make a mistake, no great harm. But if you put systems into production in finance, medicine, transportation, and so on, the errors translate into victims, and the errors become chaos.

"We have these black boxes that give answers."⁷

Today, after a good number of years, we are in a not too dissimilar situation. We are unprepared. Boxes of dark gray tending to smoke black. Independently of my chromatographic analysis of how little is understood of this machine, on Saturday SeneQCa will vote as a full member of the cabinet for the first time. This action could lead to a change in political equilibria where there is a computer as a political actor.

In the versions of the machine and in the ways we operated her, SeneQCa at first supplied only numbers to interpret. Nothing more than Monte Carlo simulations. Nothing beyond game theory simulations. Then SeneQCa began to give opinions, many of which turned out to trace common sense and the general sentiment of the people. In my view he does not even read the papers anymore.

I would be curious to know how much useful data he ingests today. The fact is that he says nothing extraordinary or against the current. He says predictable things. Commonplaces. He seems to me a monotonous computer. His strong point is discerning a minimum of signal from a tsunami of extremely noisy information. Too much stuff gets written, soaked in contradictory ideas, difficult to filter. There, SeneQCa, however much they say he solves enormous operations research problems, or does who knows

⁷Michael I. Jordan, On the pitfalls of Machine Learning, June 2022, Simons Institute for the Theory of Computing.

what in artificial intelligence, is not a computer, he is a filter: he pulls out information useful for making sensible decisions. An amplifier of the obvious.

Of course, computers can provide a statistical analysis usable by human political decision makers. They can whisper data-based opinions into your ear. Are these opinions advice? And if they were, then would computers be able to assist or even replace human decision makers? With SeneQCa, even if we are talking about an experiment, we are going in that direction, with results of clear impact on people's lives.

Intelligence is not properly the capacity to solve problems. An algorithm solves a specific problem. Solving systems of linear equations is solving a problem just as doing additions is. An algorithm is a mechanical procedure. The intelligent one is not whoever implements it but whoever discovers it. If I am handed a list of actions written on a sheet of paper and given the input of a problem, then I will solve it by faithfully following the instructions. This does not demonstrate my intelligence. It demonstrates specifically my capacity to follow instructions with precision, something that unfortunately I do not do well and have never known how to do.

The Chinese room, or Searle's room, is a thought experiment proposed by the philosopher John Searle⁸ in 1980, and it connects well to this idea. The experiment requires a person shut in a room. Sheets of paper with symbols are passed under the door. The person in the room has a book that describes rules for forming sequences with the symbols on the sheets. For example, the symbols are Chinese ideograms and the rules of the book say how to form sentences, which are then pushed out of the room under the door. Even if the sentences are correct, does the person in the room speak Chinese? The experiment is traditionally used as an argument to show that computers do nothing but execute programs, that is, apply rules to symbols, and do not understand what they are really doing.

Giving an intelligent definition of intelligence is difficult. Intelligence is the capacity to find the solution to new problems. Solving new problems can be done by adapting solutions of old problems, by using known instruments in a new way, or by inventing solutions on the basis of what one knows, often through the creativity to take shortcuts. Integrating new information and putting it together with what is already known is

⁸Searle, J. (1980). Minds, brains, and programs. *Behavioral and Brain Sciences*, 3(3), 417-424.

important. Building new information through abstract thought is sometimes necessary. Integrating and building require, in any case, thought and adaptability.

Isaac Newton watches the apple fall and arrives at the law of universal gravitation. August Kekulé dreams of a snake eating its own tail and arrives at the structure of benzene.

Intelligence, then, is connecting objects and ideas in unexpected ways. And this can refer also to the way we behave in a certain circumstance on the basis of what we manage to perceive in contact with other people. They call it emotional intelligence, in any case, always considering the solution of a problem.

The fewer resources we use for the solution, the more intelligence there is, quantitatively speaking. Solving in the most efficient way possible a problem for which no way of solving it was known demonstrates intelligence.

Creativity, then, is the instrument for arriving at the method of solution. Sometimes that method jumps out of the black box. See Newton. In this, SeneQCa and the more poetic Homo cogitans, if you please, seem similar. People are probably not conscious of their creativity. It is their brain that acts for them when it pleases and how it pleases. Sometimes it finds shorter roads through obscure processes, going to stir among memories known and never consciously seen. Then it gives you the solution while you believe you have thought something, given your intelligence. You are only the vehicle for transporting it, that brain, almost as if it were a parasite. Am I dehumanizing us? No.

Is the computer human, all too human, or is the human being computer, all too computer? In 1950, a little more than a hundred years ago, Alan Turing published a scientific article titled *Computing Machinery and Intelligence*⁹. At the center of the article was an experiment to understand whether a machine is intelligent. It is called the Turing test. It is beyond famous.

According to the experiment, a machine is intelligent when it is capable of having a written conversation with a human being and the human being cannot tell whether he is talking with a machine or with another person. It does not convince me. To catch that computer out you have to ask difficult questions. They require the machine to reason, to maneuver with language and context. Like: “Can you tell me a joke that makes you

⁹A. M. Turing (1950) *Computing Machinery and Intelligence*. *Mind* 49: 433-460.

laugh?” It is enough to have in memory a list of guaranteed jokes. “Can you tell me about a particularly meaningful experience?” Same thing. Just as I am not convinced by the approach sometimes traced back to late nineteenth century words belonging to the Countess of Lovelace: “computers have a mind only when they can create new things.”¹⁰

Computers create new things all the time today. They originate a lot of stuff. They paint. They compose catchy symphonies. They write librettos for the theater. They program themselves. From Saturday they will even be political animals. Do they assemble things already seen, with a little randomness, or do they create? Can we answer this question for people? How does one determine whether a person is really creating something new or putting together a hydra of other people’s thoughts? Think of a new animal, never seen. What do you think of? A hybrid of many different animals, already seen. An old example from the philosopher Hume. Do machines have a mind only when we do not understand how they managed to create an image? We function exactly like the machines.

“How do you invent these incredible sentences?” I asked a well known writer of theatrical monologues. He answered: “The words arrive like meteors. I drug myself with the saliva I produce while I speak.” The brain is a black box, at best a gray one, but not much lighter than SeneQCa and all these artificial intelligence instruments by now in routine use. It works and we do not know why. The idea of intelligence should be reformulated, and it is admissible that under the new definition we would turn out to be absolutely a-intelligent. Just as well.

I get out of the car. I close the door behind me. I cross the walkway in front of the hotel entrance. I put the phone in my pocket and pull it back out twice. Damned little machine. I look at the message again: *the struggle of maturity is to recover the seriousness of a child at play, friedrich nietzsche*. I answer on impulse, in English as well: “Are you a truth seeker? A friend.” I press send. My message slips out of the phone with a sound similar to how you say fish in Mandarin Chinese. *Yú*.

Six, seven years ago

¹⁰Bringsjord, S., Bello, P. & Ferrucci, D. Creativity, the Turing Test, and the (Better) Lovelace Test. *Minds and Machines* 11, 3–27 (2001).

We were told of a billionaire interested in investing capital in the company. He lived, or had a property, in Maui. I was the one who had to meet him. I have forgotten the reason. Probably Melvin was busy, because always busy. The billionaire was extremely kind in paying for my trip and hosting me at his house. I had never been to the Hawaiian islands. As always I took my little backpack and went to the airport with maximum pragmatism, exactly the way we enter a swimming pool whose water is less warm than we would like: with a dive, without wetting wrists and ankles.

I have two, three interesting memories of that trip. Some points of my account are more novelized than others, and reality, however hard I have tried to describe rather than narrate, or to embellish the discourse with personal details, has been filtered all the same.

The billionaire, I will give him an invented name, is called Leslie. In his interaction with me, Leslie has always been an excellent, impeccable person, so I do not feel like criticizing, even if my account, at moments, will seem to use a little sarcasm. It will be an involuntary use. After these years, Leslie would appreciate my words, as a teacher can appreciate the old compositions of a former pupil by now getting on in years.

I arrived in Maui at night, after a trip made, as always, in economy class, seated at the back of the plane. Two members of his staff came to pick me up at the airport in a big black car with gleaming studs. It had tinted windows. Leather seats. There was fog and little light along the road. Once the gate on the drive opened, the driver kept going for a minute before stopping. I would lodge in an annex, in a beautiful spot facing the Pacific, something I realized only on waking.

They welcomed me without formality. I would sleep in a little house, ground to roof mine, which I could reach by walking along a path lit by photovoltaic spotlights. There was no ostentation. There was order. Sobriety. He was the first billionaire I had interacted with, so I did not quite know what to expect. I was accompanied to the door of my room by an elderly lady, extremely affable, with a smile printed on her face, but a natural one. When she handed me the key, she made an expression of satisfaction, almost as if to tell me that there I would find a small refuge from the rest of the world.

“The Wi-Fi password?” I asked.

“There isn’t one.”

“There is no password?”

“There is no Wi-Fi at all. In fact, here not even the phone gets signal. Mister Leslie is convinced that radio waves are bad for the health.” Fine, then. I passed some calm days after the first hours of panic.

When I entered the room, I was struck. The room was full of xylophones. Full. They were everywhere. The room was a museum of xylophones. Some rested on metal stands, others were on wobbly shelves or directly on the floor, as if they had been there for years in a strange collective meditation. There was space to reach the bed from the door. There was a gray couch and an empty zone to be used as a walking lane from the couch to the bed and from the couch to the adjoining bathroom.

Every instrument seemed to have a personality of its own, some with wooden bars, opaque and lived-in, others in shining metal, impeccable, as if awaiting a concert that was due to begin shortly.

The metal bars materialized in a surreally sunset light, projected by an orange lampshade. The room was a conspiracy of normality and absurdity that my paranoia as an exhausted traveler could not decipher. The bars struck me at once as in a brief psychedelic nightmare made of parallel objects, like the rungs of a ladder, the slats of a shutter, the grill of a barbecue, or the bathroom rack for towels. I did not see musical instruments, but maybe from the weariness of the trip, and with jet lag drilling into my brain, I saw only horizontal bars, ordered with geometric precision, in a totally artificial and oppressive visual harmony. The xylophones were passive instruments of torture. They did not touch me, they did not interact with me, and yet their mere presence, that implacable geometry, was enough to bother me. And I wondered whether I would be able to sleep surrounded by that claustrophobic order.

I moved with caution, avoiding all contact. The xylophones followed a scheme in their order: the smallest dominated the center of the room, while the big ones were confined to the corners, like flunked children, shy, during a school play. I touched them a couple of times with the knuckle of my index finger, leaning forward. The sound that came out was deep, and it carried with it a sensation of peace that almost made me ashamed of my negative emotions. I went to bed forgetting that arrangement and the strangeness of the place. It was a very comfortable bed. I slept serenely under the effect of melatonin. In sleep an unexpected gift arrived. I dreamed of an old friend. We were sitting in a provincial bar. The sun was shining outside the windows, and we were

drinking a coffee as if all those years had never passed. I heard from far away the bells of the church of San Lorenzo.

At exactly 6 in the morning, a group of birds began to sing. It felt like being in the tropical forest. Different songs lasting a few minutes. I said to myself, “Listen to these birds. They all sing together at 6 in the morning.” The next day, the same thing. And so the third day. I thought, “Is it possible that these pests go off daily at 6? Incredible.” I later discovered that this was the alarm clock. A recording that changes randomly every day and grows in progression. The sound came from loudspeakers positioned scattered around the garden, partly to give the sense of three-dimensionality in nature, where the birds do not all sit on the same branch, but in different places in the wood. They had designed an alarm as natural as possible, which, in its artificiality, turned out incredibly invasive. At least, I perceived it that way.

I met Leslie on the third day. The billionaire had facial skin that was incredibly glossy, almost luminescent. Tall and slender, he radiated ease. At the time, his approach to life revolved around an unusual conviction. He firmly believed that, by following well defined practices, it was possible to live to 300 or even 400 years. First among them all, drinking special water. By the liter. While I was in the company of a member of his staff, I was thirsty, and in one of the kitchens I opened a tap to drink a glass of water.

“If you want better water, take that one,” and I was pointed to a big light blue plastic bottle.

“It is glacier water.” I wondered whether it was drinkable. As I was about to open the big bottle, the person in the room added, “If you want even better water, drink this other one.” I was surprised.

At the back of the room there was a glass bubble with a vortex inside, green moss, and violet lights. “Volcanic water, alkaline and ionized. It guarantees maximum hydration.”

I thanked them and gulped it down, a little the way one drinks a magic potion. The perception of my health improved between the first and the second sip.

On the billionaire’s food a special discussion is needed. He believed strongly in the beneficial power of mushrooms. He drank dark, dense fluids containing mushrooms, herbs, and acorns. By now around seventy, he had the body of a twenty year old. I, who was younger, felt flabby and stooped. A team of mycologists dictated his diet. He told me

how it took years to find the ideal one and the right doses of cordyceps militaris. Almost lethal errors were made. Perhaps the most singular and frightening was when, having taken a wrong quantity of yartsa gunbu, a mushroom known for its strong aphrodisiac properties, he was afflicted with priapism for several days. The timely intervention of traditional medicine was needed to bring the situation back to normal. The wagging tongues insinuate that he has been condemned to a perpetual half erection, in aeternum, like a curious joke of fate.

Wagging tongues or not, Leslie signed a document produced by his lawyers committing to give us fifty million, for a project up in the air that, honestly, not even I was able to explain. As soon as he learned of it, Melvin launched into the Lambada. Dance and humming included.

In my conversations with Leslie I kept pointing out the problems of the project. When I should have been selling, I spoke only of obstacles. He answered: “Sometimes it is right to invest in dreams.” In his wake other investors came in, until we raised a crazy quantity of capital in a totally unjustified way. They competed to talk with us.

We walked barefoot on the beach. I carried my winter shoes with thumb and index finger, while he had long, almost webbed feet. At sunset hour, Leslie observed: “You see, Dario, I know two truths. The formal one, of mathematics, which people like you reach through a proof. Right?”

I nodded, without understanding whether he wanted to force my arm behind my back, figuratively, to “prove” his intelligence to me with a billionaire’s speech. The use of the verb “to know” pleased me. He did not say “there are.”

Then he continued: “There is another truth, though, for me, and I am sure you will not accept it. And that changes nothing between us. It is the truth supplied to us by what we believe, closer to our emotions than to our reason, it is a truth we decide to believe in. Dario, the diversity in our thoughts is our strength.”

He explained to me that in Sanskrit truth is said vrata and has a meaning of its own. “The vrata is a personal commitment, a pact one makes first of all with oneself. In Jainism, for example, it is a devotion, not imposed from outside but intimately chosen. Here truth is not an objective correspondence with the facts, but an alignment between what we promise ourselves and what we do. An inner coherence that becomes sacred precisely because it is free.”

He spoke like a written book. I was disarmed, because what he said had a certain fascination. It was bewitching. I tried to go straight to the point, letting slip a moment of authenticity, perhaps one to avoid: “Do you really believe in the project? Because I, to tell the whole truth, am not sure.”

He smiled and said: “Vrata. I believe in you all, not necessarily in what you do. Something good will come out. Freedom leads to harmony.”

This unexpected trust of his made me feel closer to him, so much so that I found myself on the point of revealing too many things, without a reason: technical details on which I could not have sustained a deep conversation, doubts that would have shown my indecision, or simply an extremist’s pessimism about the project. I limited myself to speaking vaguely, luckily. After all, this remained a professional relationship, and he was an investor, a person in a position of control. One wrong sentence, a moment of excessive confidence, or a negative emotion involuntarily aroused could have erased everything. My insecurity about the possibility of making a mistake was greater than my insecurity about the project. However sincere Leslie seemed, it was right not to trust, or to try to predict his reaction. I spoke:

“Leslie, you are a person with great experience. I imagine you have seen all kinds of things in your career. Maybe it is that I have impostor syndrome. You know, all this capital. I am not sure. The company. The future. The computer, I do not believe it will ever work...”

Without changing expression, as if those words were already familiar to him, as if he had heard them a hundred times and knew exactly what to answer, he spoke with a surprising calm. Once again I was struck by the precision with which he articulated his sentences, almost as if every word had been chosen with care in advance. The glossiness of his skin was a metaphor for the linearity with which he spoke.

“Every failure has taught me something, always. It has been a way to learn, to grow, even when it did not seem so. Your sense of inadequacy, Dario, is nothing but the reflection of your intelligence. Only the stupid live free of doubts, blind before their limits. But what really counts, what makes the difference, is to move. Not to stand still. This is what distinguishes those who try, those who attempt, from those who give up. It is never a question of not making mistakes, but of continuing to search, to take one step after another. The important thing, Dario, is to move.”

He paused, then added a hint of a smile:

“But you already know that, no? And if you do not know it, you will find out soon.”

“Thank you. I appreciate it,” was all I said, essentially to cut short a conversation that was better closed there. He went on, by now wound up like a music box, surprising me even more with the content.

“You know what I see in this whole venture capital circus, Dario? A great elaborate ritual. Like the rain dances, only here we dance with money. One venture puts in capital, then another, then another still. Not because they really believe in the product, but because they believe in the dance itself. In the fact that someone else will believe even more than they do. And in the end they will make money. All a question of faith and convictions. The important thing is that the valuation of the company grows.”

He stopped to look me in the eyes. Without intensity or pressure, the way one looks at a friend:

“It is all a great play of mirrors. The company? The product? Details. What you sell could be fried air or the cure for cancer, it makes no difference. Sadly. The important thing is that the music continues, that the next investor enters the dance. Everyone knows it. It is a Ponzi scheme. You cannot rule out, though, that from this scheme some good may also emerge, in the form of jobs, in the form of innovation, and in the form of lessons for the future.”

He took off his T-shirt and shorts, the kind that reach the knee, almost knickerbockers. He remained naked, and before I could turn the other way, I gathered that for him nudity must be a fairly natural thing.

He immersed himself in the ocean up to his shoulders with the same nonchalance, while one of his bodyguards with sunglasses watched us from very far away, perched on a quad bike in a corner of the villa's garden. I could not help noticing that, as he headed long-limbed toward the water, the setting sun created a metallic reflection on the polished outside of his left buttock. I thought again of the xylophones in my room.

When I said goodbye, thanking him profusely, he gave me a cream of porcini and morels to be used as a face mask or aftersun cream. I confess that one evening, just back in London, I tried using it. It had a good smell.